"Just One More Clip":
Short Videos, Big Self-Control Problems*

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Abstract

I study how short-form design amplifies self-control problems in digital media. Short units repeatedly reactivate temptation that lasts longer than each unit of content, turning local temptation into sustained overconsumption. Using microdata from a U.S. short-drama platform, I exploit a nonlinear top-up menu to infer viewing plans and show that paying users watch 82.1% more than intended. Structural estimates imply an average temptation horizon of 11.2 minutes, short relative to the full drama but long relative to one-minute episodes. Larger bundles, default time limits, and breaks improve consumer welfare. A short-video calibration highlights the broader welfare relevance of short-form design.

JEL codes: D12, D91, I31, L82

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1 Introduction

Digital platforms increasingly organize consumption as a sequence of small continuation decisions: one more clip, one more episode, one more scroll, one more round. This design changes the economics of self-control. A consumer who would reject a long commitment may repeatedly accept a short one. When short-run temptation lasts longer than the unit of consumption, each stopping decision arrives while the next unit is still locally tempting. The result is a rolling "present": users plan to stop soon, but the stopping point keeps moving forward.

This mechanism is relevant for current policy and legal debates over digital platform design. Policy responses now range from platform-level screen-time defaults and break reminders to age-based restrictions on social media access.[1] In *K.G.M. v. Meta et al.*, a recent bellwether case in Los Angeles Superior Court, a jury found Meta and Google negligent in the design or operation of products alleged to induce compulsive use.[2] The case does not establish the economic mechanism studied in this paper. It does, however, sharpen the relevant economic question: when do platform-design choices generate welfare losses rather than merely satisfy high demand?

The empirical challenge has two parts. The first is to distinguish heavy usage from distorted usage. High engagement alone is not evidence of a welfare problem, since consumers may simply value the content. The second is to recover heterogeneity in planned continuation. This heterogeneity is central because the welfare consequences of short content depend not only on the magnitude of temptation, but also on how long temptation lasts and how this duration varies across users. Identifying the distribution of temptation horizons therefore requires observing more than whether users continue; it requires variation in the length of continuation plans.

I study these questions using microdata from a U.S. short-drama platform. Short dramas are serialized mobile videos divided into many brief episodes, typically around one minute each. In the main user-level data, the focal drama has 80 one-minute episodes. The first nine episodes are free; subsequent episodes must be unlocked sequentially with platform tokens. Users buy tokens through a nonlinear top-up menu with four packages at different price points: larger packages offer lower prices per episode but require larger upfront payments. This menu provides the paper's main revealed-preference variation. A larger package implies a longer package-implied continuation plan at the purchase moment, while a smaller package implies a shorter one. Repeated purchases of small packages despite lower unit prices for larger packages reveal failures to implement short continuation plans. At the same time, the availability and positive market shares of multiple package sizes provide distributional variation in planned continuation, which helps identify the distribution of temptation durations.

The reduced-form evidence from the focal drama shows that the gap between package-implied plans and realized viewing is large, heterogenous, and persistent. Among users who purchase tokens, the first top-up choice implies planned viewing of 28 minutes, while realized viewing av-

[1] For example, TikTok introduced a default 60-minute daily screen-time limit for accounts held by users under age 18, while Australia adopted age-based social-media restrictions for users under 16.

[2] The companies announced plans to appeal the verdict. TikTok and Snap, which were also defendants, settled before trial. See Reuters (2026); Huamani and Ortutay (2026); Crowell & Moring LLP (2026).

erages 51 minutes. Realized viewing is therefore 82.1 percent higher than the viewing implied by the first purchase. Many users also pay more than necessary for the episodes they ultimately unlock: 39.1 percent of paying users spend more than the ex post cost-minimizing package sequence, with average overpayment of $5.51. These patterns are not confined to first purchases. Users repeatedly return to the smallest package after substantial viewing experience, and the probability of buying the smallest package again rises with the number of previous small-package purchases. The full distribution of package choices is also informative: all four packages have positive market shares, with many users choosing short implied continuation plans while others choose larger commitments. This variation is what allows the structural model to discipline heterogeneity in temptation duration, rather than only its average effect.

This persistence is difficult to reconcile with a learning-only model. Under Bayesian learning about drama quality, small initial purchases can be rational: users may buy a small package to experiment. As experience accumulates, however, uncertainty should fall, and belief updating should sort users toward either larger packages after favorable signals or stopping after unfavorable signals. The data show a different pattern. Users become increasingly likely to make the same low-commitment purchase again. This pattern runs against the martingale property of rational learning, under which users' current beliefs are already their best forecast of future beliefs and do not systematically drift toward repeated small trials. The evidence instead points to a rolling local-repurchase force rather than experimentation alone.

The content-length mechanism is also visible beyond the focal drama. Across 353 dramas on the same platform, shorter episodes are associated with higher completion rates: in the full specification, an additional minute of episode length predicts a 2.42-percentage-point lower completion rate, equal to 21.2 percent of the mean. This evidence is descriptive, but it shows that the content-length margin emphasized by the model appears in platform-wide data.

I formalize the mechanism in a stylized model of sequential media consumption. Users derive welfare-relevant utility from content but also experience short-run temptation over a finite horizon. When the content unit is shorter than this horizon, the next unit remains tempting at each stopping decision. A naïve user plans to stop after the currently tempting units, but when that future decision arrives, temptation has shifted forward. The model delivers the paper's main comparative static: shorter content exposes more users to self-control problems and increases temptation-induced surplus losses under a long-run welfare criterion. It also clarifies why commitment-like design can help: longer content units, default time limits, and mandatory breaks all work by weakening the repeated renewal of local temptation.

I then estimate a dynamic model of token purchases and episode unlocking. Users differ in their value for the drama and in the duration of temptation. They learn about drama quality, accumulate habit within a viewing round, hold token balances, and choose whether to purchase tokens or stop. Welfare utility includes content value, outside options, habit, and monetary costs. Choice utility additionally includes a local temptation term. The key parameter is the temptation horizon: the number of minutes over which immediate continuation is overweighted.

The nonlinear top-up menu disciplines this horizon through package choices and subsequent transitions. Package-size choices provide information about planned continuation and heterogeneity in temptation duration: users with longer temptation horizons anticipate more continuation and are more likely to choose larger packages. Repeated small-package transitions provide especially direct evidence on the renewal of local temptation. Transitions after early small-package purchases help distinguish temptation from learning: a learning-only model predicts sorting toward upgrading or stopping as uncertainty resolves, whereas renewed temptation predicts persistent local repurchasing. Habit formation is disciplined by state dependence in continuation and top-up decisions within viewing rounds.

The estimates show that effective rolling temptation is short in clock time but long relative to the unit of consumption. The average effective temptation horizon is 11.2 minutes. This is shorter than the full 80-minute drama, but much longer than a one-minute episode, so temptation remains active across many consecutive episode-level decisions. The estimated temptation magnitude is 23.3¢ per minute, which is economically large relative to the estimated per-episode drama value, which averages −2.2¢. The model also finds substantial heterogeneity: 28.8 percent of users have zero effective rolling temptation, meaning that their observed choices are not distorted by the rolling-temptation channel. This mass may include users who experience no temptation and sophisticated users who anticipate and neutralize future temptation. Conditional on positive effective temptation, durations are drawn from a heavy-tailed distribution. Learning is rapid. By the first paid top-up decision, posterior uncertainty about drama quality has fallen by 87 percent, to 1.3¢. Repeated small-package purchases are therefore not primarily driven by slow resolution of uncertainty. Habit formation is also present: each additional episode watched within a round raises subsequent per-episode utility by 1.6¢ on average. Unlike temptation, however, habit formation raises welfare-relevant utility rather than distorting perceived short-run value.

Under the long-run welfare criterion used in the model, the estimated self-control component generates economically meaningful surplus losses. Average user surplus is −$0.25 in the baseline model. Eliminating temptation raises average user surplus to $0.58, implying a temptation-induced loss of $0.83 per user. These losses are concentrated among users near the margin of valuing the drama: temptation induces them to continue purchasing and watching even though their welfare-relevant continuation value is low. This concentration is important. The paper does not argue that all high engagement is harmful. It shows that short-form design has the largest welfare effects for marginal users whose realized continuation exceeds their own package-implied plans.

Counterfactuals show how design changes affect this margin. Default time limits and mandatory breaks increase user surplus by weakening the repeated renewal of local temptation. A default time limit tied to the first top-up choice reduces the welfare loss from temptation by 37.3 percent when opting out requires a modest cost. A five-minute mandatory break before top-up reduces the loss by 21.7 percent. Longer content units operate through the same mechanism: increasing paid episode length from one to eight minutes reduces temptation-induced surplus losses by 12.4 percent. I also study removing the smallest package as a mechanism counterfactual. By reducing the low-commitment repurchase margin, this intervention lowers the welfare loss from temptation by 55.4 percent. Across these exercises, the common lesson is that the effective size and frequency of stopping decisions shape welfare when temptation renews.

Finally, I use the estimated temptation structure to calibrate welfare stakes in a broader short-video environment.[3] Under a full-transfer calibration, eliminating temptation raises hourly consumer surplus by \$2.16. Aggregating this quantity using 170 million U.S. TikTok users and 29.2 monthly hours implies a monthly temptation-induced consumer-surplus loss of \$13.3 billion. A 12-minute content-length counterfactual recovers \$7.9 billion per month, while a 10-minute default time limit recovers \$7.8 billion. These counterfactuals remain total-welfare-improving unless platform plus advertiser surplus exceeds \$6.56 and \$5.19 per viewing hour, respectively. I also relax the transfer assumptions along several margins, allowing for partial population comparability, weaker temptation magnitude, shorter temptation horizons, alternative value distributions, and imperfect default compliance. The magnitudes remain economically meaningful. These calculations should be read as scale benchmarks, but they show that the interaction between temptation duration and content length can have large welfare implications beyond the short-drama setting.

Contribution and related literature. The paper makes three contributions. First, it contributes to the literature on dynamic inconsistency, temptation, and commitment by making the duration of temptation an estimable object. Starting with Strotz (1955) and Pollak (1968), this literature studies how current choices can conflict with future plans (Laibson, 1997; O'Donoghue and Rabin, 1999; Gul and Pesendorfer, 2001, 2004). Empirical work has identified self-control problems using commitment devices, costly overpayment, and experimental choices (DellaVigna and Malmendier, 2004, 2006; Gruber and Köszegi, 2001; Giné et al., 2010; Augenblick and Rabin, 2019). Relative to this work, I estimate not only whether behavior is consistent with self-control problems, but also the horizon over which short-run temptation affects choices. The estimate is intuitive: temptation lasts minutes, not hours, but minutes are enough when content is divided into one-minute units.

Second, the paper contributes to work on digital addiction by providing a revealed-preference approach to measuring package-implied planned viewing. A growing literature estimates the welfare and well-being effects of digital media use (Allcott, Braghieri, Eichmeyer, and Gentzkow, 2020; Allcott, Gentzkow, and Song, 2022; Braghieri, Levy, and Makarin, 2022; Mosquera, Odunowo, McNamara, Guo, and Petrie, 2020; Collis and Eggers, 2022), studies self-control tools in smartphone use (Hoong, 2021), and analyzes collective traps, peer effects, and addictive platform design (Bursztyn, Handel, Jimenez, and Roth, 2023; Barwick, Chen, Fu, and Li, 2024; Ichihashi and Kim, 2023; Aridor, Jiménez-Durán, Levy, and Song, 2024; Beknazar-Yuzbashev, Jiménez-Durán, and Stalinski, 2024; Gao, Gao, Meng, and Yu, 2024). My contribution is to bring real monetary continuation choices to this setting. The nonlinear top-up menu generates a model-based measure of package-implied planned viewing, allowing me to compare planned and realized consumption within the same digital environment. This comparison distinguishes high willingness to pay from temptation-driven continuation using actual purchases rather than stated intentions or usage alone.

[3] This exercise is not an estimate of welfare on TikTok or any specific platform: unpaid algorithmic feeds differ from paid short dramas in pricing, content supply, social interaction, and outside options.

Third, the paper shows how product granularity shapes welfare. The same temptation horizon can have different consequences depending on the length of the decision unit. A short horizon may be harmless when decisions require large commitments, but harmful when platforms repeatedly ask users to make small continuation choices. This connects to work on tempting goods and rationing: consumers may restrict access to vice goods to manage self-control problems (Wertenbroch, 1998), and related work studies time-inconsistent preferences in digital content (Hoch and Loewenstein, 1991; Dhar and Wertenbroch, 2000; Zhang, Chan, Luo, and Wang, 2022). This paper turns the rationing logic around. Instead of asking how consumers restrict access to tempting goods, it studies how platforms choose the granularity of access and thereby shape the frequency of stopping decisions. Related work on inertia, defaults, and contract design also shows that seemingly small design features can strongly affect consumer choices (Handel, 2013; Heiss, McFadden, Winter, Wuppermann, and Zhou, 2021; Miller, Sahni, and Strulov-Shlain, 2025).

This third contribution also connects the paper to empirical models of dynamic demand with learning, habit formation, and addiction. Models of rational addiction and habit formation explain persistence through state dependence in welfare-relevant utility (Spinnewyn, 1981; Becker and Murphy, 1988; Becker, Grossman, and Murphy, 1994; Orphanides and Zervos, 1995; Zhen, Wohlgenant, Karns, and Kaufman, 2011). Dynamic demand models with state dependence, learning, switching costs, and experience goods provide tools for distinguishing persistent tastes from changing beliefs and frictions (Rust, 1987; Shapiro, 1983; Bergemann and Välimäki, 2006; Dubé, Hitsch, Rossi, and Vitorino, 2008; Gordon and Sun, 2015; Chen and Rao, 2020; Honka, 2014; Shcherbakov, 2016). In my setting, learning, habit, and temptation all predict persistence, but they imply different top-up and unlocking patterns and have different welfare interpretations. The nonlinear menu and high-frequency consumption histories help the structural model discipline these mechanisms and evaluate their welfare content under its maintained assumptions.

The rest of the paper proceeds as follows. Section 2 presents the stylized model and derives the content-length comparative static. Section 3 describes the short-drama platform and documents the reduced-form evidence from episode length, overpayment, and repeated small-package purchases. Section 4 develops the structural model. Section 5 discusses estimation and identification. Section 6 presents the welfare analysis and counterfactual design exercises. Section 7 applies the estimates to a broader short-video environment. Section 8 concludes.

2 Stylized Model of Temptation

I use a stylized model to isolate how short-form content amplifies self-control problems through the magnitude and duration of temptation. I model temptation as an additive component of perceived utility. This formulation fits naturally with discrete, sequential video consumption, preserves the qualitative predictions of the canonical multiplicative β–δ model of present bias, and leads to a simpler estimating equation in the structural model (Banerjee and Mullainathan, 2010;

Allcott, Gentzkow, and Song, 2022).[4] The model highlights the main comparative static: when the unit of content is shorter than the duration of temptation, temptation rolls forward with consumption and generates repeated failures to stop. Section 4 embeds this mechanism in a structural model of short-drama consumption with top-up choices, learning, and habit formation.

2.1 Setup

Consider a platform offering a large set of one-minute videos, indexed by $\{1, \ldots, T\}$. Users make decisions at the video level, watch sequentially, and derive a constant intrinsic utility $x \in \mathbb{R}$ from each video. I assume for now that users observe x and defer uncertainty and learning to the structural model. The value of outside option is normalized to zero. In addition to intrinsic utility, users may experience a *temptation* to keep watching. Temptation adds $\kappa > 0$ per minute to perceived utility over a horizon of $\chi \in \{0, 1, 2, \ldots\}$ minutes. I interpret $\chi = 0$ as zero effective rolling temptation: the user's observed choices are not distorted by the rolling-temptation channel. This reduced-form temptation can be microfounded by present bias (e.g., Laibson, 1997; Gruber and Köszegi, 2001), a taste for novelty (e.g., Modirshanechi, Lin, Xu, Herzog, and Gerstner, 2025), inertia, or other behavioral mechanisms.

Let t denote the *current* video, i.e., the video at which the user is making a decision, and let $\tau \geq t$ denote an *evaluated* video, i.e., a video whose utility is assessed from the perspective of current video t. The perceived utility from video τ at current video t is

$$\tilde{u}_{\tau}^{t}(x, \chi) = \underbrace{x}_{\text{intrinsic utility}} + \underbrace{\mathbb{1}\{\tau < t + \chi\}\kappa}_{\text{temptation } \kappa \text{ with duration } \chi}. \tag{1}$$

The indicator captures the user's current effective temptation horizon. A user with $\chi = 0$ perceives the true utility x from all videos. A user with $\chi = 2$, by contrast, perceives utility $x + \kappa$ from the current and next videos, t and $t + 1$, and true utility x from all later videos. In this section, I focus on users with $x \in (-\kappa, 0)$. These users dislike the videos relative to the outside option in welfare terms, but may nevertheless watch videos that fall within their temptation horizon.

The key feature of this specification is that temptation is tied to the user's current perspective rather than to a fixed set of videos. As the user moves forward, the temptation horizon moves forward as well. The same video can therefore lie outside the temptation horizon at one decision point but inside it at the next. Table 1a illustrates this logic for users with $\chi = 2$. At $t = 1$, video 3 lies outside the temptation horizon and is valued at $\tilde{u}_3^1 = x$. After the user reaches $t = 2$, video 3 becomes the next video and enters the temptation horizon, so its perceived utility rises to $\tilde{u}_3^2 = x + \kappa$. Temptation therefore rolls forward with consumption: the extra perceived value attached to the near future is repeatedly renewed as each subsequent video becomes immediate.

[4] Appendix A.1 shows how the perceived utility in equation (1) can be represented using quasi-hyperbolic discounting, which microfounds the temptation term.

Table 1: Example: Self-control problems with $\chi = 2$ and $x \in (-\kappa, -\kappa/2)$

(a) Baseline

Evaluated video τ	Current video t: 1	2	3	4
1	$\boxed{x+\kappa}$			
2	$\boxed{x+\kappa}$	$\boxed{x+\kappa}$		
3	x	$\boxed{x+\kappa}$	$\boxed{x+\kappa}$	
4	x	x	$\boxed{x+\kappa}$	$\boxed{x+\kappa}$
5	x	x	x	$\boxed{x+\kappa}$
6	x	x	x	x

(b) Comparative static: Video length $n = 4$

Evaluated video τ	Current video t: 1	2	3	4
1	$4x + 2\kappa$			
2	$4x$	$4x + 2\kappa$		
3	$4x$	$4x$	$4x + 2\kappa$	
4	$4x$	$4x$	$4x$	$4x + 2\kappa$
5	$4x$	$4x$	$4x$	$4x$
6	$4x$	$4x$	$4x$	$4x$

Notes: The tables illustrate a naïve user with intrinsic value $x \in (-\kappa, -\kappa/2)$ and temptation duration $\chi = 2$. Each entry reports the perceived utility $\tilde{u}_\tau^t$ from watching video τ from the perspective of current video t. Columns correspond to current videos, and rows correspond to evaluated videos. Positive utilities are boxed; highlighted boxes indicate the videos the user actually watches. Panel (a) shows the one-minute baseline, in which the user plans to watch two videos at each decision point but ultimately watches all available videos. Panel (b) shows the case with four-minute videos, in which the perceived flow utility from starting a video is $4x + 2\kappa < 0$, so the user chooses not to watch.

2.2 Results

The stylized model generates self-control problems through the rolling-forward nature of temptation. Consider a user with $x \in (-\kappa, 0)$ and temptation duration $\chi \geq 1$. At current video t, the χ videos inside the current temptation horizon have perceived utility $\tilde{u}_\tau^t = x + \kappa > 0$, so the user plans to watch them. She expects to stop afterward, when perceived utility returns to $x < 0$. Once she reaches video $t + \chi$, however, the temptation horizon shifts forward, and the next χ videos again become tempting. The user therefore repeatedly revises her stopping plan and ultimately consumes all available videos. Table 1a illustrates this logic for a user with $\chi = 2$, who always plans to watch two more videos but never follows through on the planned stopping decision.

This behavior generates consumer surplus losses. In the benchmark without temptation, users watch if and only if their intrinsic value is nonnegative. Thus, users with $x < 0$ who consume because of temptation experience a welfare loss. Let (x, χ) represent a user, and let $S \subset \mathbb{R} \times \mathbb{N}_+$ denote a set of users who experience self-control problems. I define the *average surplus loss due to temptation* as

$$\Delta(S) = -\sum_{\chi \in \mathbb{N}_+} \left[T \int_{\{x:(x,\chi) \in S\}} x f_{x|\chi}(x)\, \mathrm{d}x \right] P(\chi), \tag{2}$$

where $P(\chi)$ is the probability mass of χ and $f_{x|\chi}(x)$ is the conditional density of x. Since $x < 0$ for users with self-control problems, this expression is positive.

The key comparative static concerns content length. Consider a counterfactual in which the platform groups the original one-minute videos into n-minute videos, holding total content length fixed.[5] Users still make decisions at the video level. The perceived flow utility from starting an

[5]This exercise analyzes the impact of video length on self-control problems by merging n one-minute clips into a single longer video. A real-world analogue is the U.S. law on "loosies," which prohibits the sale of individual cigarettes and mandates a minimum pack size of 20. Such a policy mitigates the effect of momentary temptation to smoke "just

n-minute video is

$$\widetilde{u}_t^t(x,\chi;n) = nx + \min\{n,\chi\}\kappa. \tag{3}$$

The intrinsic component scales with video length, while temptation applies only to the portion of the video that lies within the temptation horizon. If $n \le \chi$, the entire video is tempting. If $n > \chi$, only the first χ minutes are tempting, so the effect of temptation is diluted over the longer commitment. Given length n, a user watches if perceived flow utility is positive. A self-control problem arises when the user watches despite having negative welfare-relevant value. This occurs if and only if

$$\max\left\{-\kappa, -\frac{\chi\kappa}{n}\right\} < x < 0. \tag{4}$$

When the video is shorter than the temptation duration, $n \le \chi$, all users with $x \in (-\kappa, 0)$ remain affected. When the video is longer than the temptation duration, $n > \chi$, users with sufficiently low intrinsic value, $x < -\chi\kappa/n$, choose not to watch. As n increases, this cutoff approaches zero, so the set of users with self-control problems shrinks.

Let S_n^N denote the set of users with self-control problems when video length is n, characterized by condition (4), and let $\Delta_n^N := \Delta(S_n^N)$ denote the associated average surplus loss. The baseline surplus loss is $\Delta^* := \Delta_1^N$. Proposition 1 summarizes the comparative static.

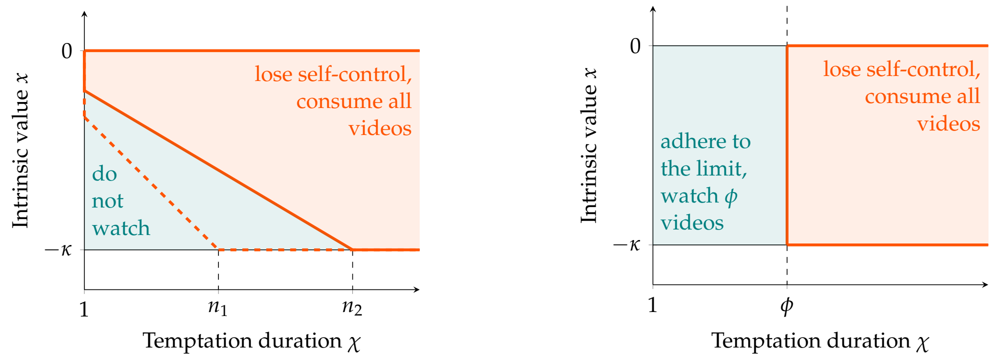
Proposition 1 (Content length and self-control problems) *For videos of length $n \in \mathbb{N}_+$, users experience self-control problems if and only if condition (4) holds. Hence, one-minute videos affect all users with $x \in (-\kappa, 0)$ and $\chi \ge 1$. Moreover, if $1 \le n_1 < n_2$, then $S_{n_1}^N \supseteq S_{n_2}^N$ and $\Delta_{n_1}^N \ge \Delta_{n_2}^N$. Shorter videos therefore expose weakly more users to self-control problems and generate weakly larger surplus losses.*

Table 1b illustrates the mechanism for $\chi = 2$ and $n = 4$: users with $x < -\kappa/2$ avoid watching and therefore escape the self-control problem. Figure 1a illustrates the same comparative static. The orange region represents users who watch all available content despite having negative intrinsic value. As video length increases, the teal region expands: more users choose not to start watching because the longer commitment forces them to evaluate content beyond their current temptation horizon.

I next consider a policy that imposes a default screen-time limit of ϕ minutes, allowing users to opt out ex ante but not ex post.[6] To break ties, I assume users pay an arbitrarily small cost ε to opt out. Figure 1b illustrates the response. Among users with $x \in (-\kappa, 0)$, those with $\chi \le \phi$ adhere to the limit because, at the time of the opt-out decision, they do not plan to watch beyond ϕ minutes. Users with $\chi > \phi$ expect to watch more than the limit and therefore opt out. The policy acts as a commitment device: it constrains overconsumption among users with relatively short temptation durations while allowing users with longer planned consumption to opt out.

one more" cigarette and forces consumers to evaluate a larger unit of consumption.

[6] A similar policy was implemented by TikTok in 2023, setting a default 60-minute daily limit for users under age 18, with the option to opt out.



(a) Longer video length n

(b) Default time limit of ϕ minutes

Figure 1: Illustration: User behavior under counterfactual scenarios

Notes: This figure illustrates user behavior under counterfactual scenarios, where users are characterized by intrinsic value x and temptation duration χ. Panel (a) presents the comparative static for video length n. Users with self-control problems, shown in the orange region, watch all available content despite having negative intrinsic value, whereas users in the teal region choose not to watch. Panel (b) shows behavior under a default screen-time limit of ϕ minutes. The orange region represents users who opt out of the limit and continue watching all available content, whereas the teal region represents users who adhere to the limit and watch fewer videos.

Given this behavior, the average surplus loss under a default time limit is

$$\Delta_{\phi}^{TL} := -\sum_{1\leq\chi\leq\phi}\left[\phi\int_{-\kappa}^{0} x f_{x|\chi}(x)\,\mathrm{d}x\right]P(\chi) - \sum_{\chi>\phi}\left[T\int_{-\kappa}^{0} x f_{x|\chi}(x)\,\mathrm{d}x\right]P(\chi), \tag{5}$$

where users with $1 \leq \chi \leq \phi$ consume only ϕ minutes, while users with $\chi > \phi$ opt out and consume all T minutes. The baseline corresponds to either $\phi = 0$, in which case no affected users comply, or $\phi = T$, in which case the limit is nonbinding. Thus $\Delta_0^{TL} = \Delta_T^{TL} = \Delta^*$. For any interior limit $\phi \in (0, T)$, the reduction in surplus loss is

$$\Delta^* - \Delta_{\phi}^{TL} = \sum_{1\leq\chi\leq\phi}(T-\phi)\left[-\int_{-\kappa}^{0} x f_{x|\chi}(x)\,\mathrm{d}x\right]P(\chi) > 0. \tag{6}$$

The policy therefore involves a trade-off. A higher limit expands policy coverage, since more users satisfy $\chi \leq \phi$, but it reduces the surplus gain per complier because each complier consumes more content before the limit binds.

Proposition 2 (Default time limits reduce self-control problems) *For users with self-control problems characterized in Proposition 1, a default time limit of $\phi \in (0, T)$ reduces consumption to ϕ minutes for users with $\chi \leq \phi$, thereby increasing consumer surplus. The optimal limit maximizes the surplus gain in equation (6), balancing broader policy coverage against a smaller surplus gain per complier.*

2.3 Discussion

The stylized model is designed to isolate one mechanism: the interaction between content length and the duration of temptation. The contribution is to make the *duration* of temptation an explicit economic object. Interpreted through the lens of present bias, this duration corresponds to the length of the "present" over which immediate consumption is overweighted.[7] The model shows that content granularity matters because the decision unit may be shorter than this temptation horizon. In that case, each stopping decision arrives while the next unit is still tempting, so temptation rolls forward with consumption. The structural analysis later estimates the distribution of temptation durations using variation from the nonlinear top-up menu, allowing me to quantify how this distribution interacts with one-minute content.

The temptation term should be interpreted as a reduced-form representation of a short-run behavioral distortion. It can be microfounded by present bias, a temptation preference, cue-triggered addiction, a taste for novelty, inertia, or other mechanisms that raise the perceived value of immediate continuation relative to longer-run utility. The empirical analysis does not attempt to distinguish among these psychological foundations. What matters for the mechanism is that the extra perceived value is local and shifts forward as the user continues watching. These foundations are therefore behaviorally equivalent for the purposes of the stylized model: they all generate repeated local continuation despite negative welfare-relevant value.

The stylized analysis focuses on users with positive effective temptation who do not fully internalize that the temptation horizon will shift forward as they continue watching. At any current video t, these users form plans using their current perceived preferences. They therefore plan to stop after a small number of videos but revise that plan when the stopping point arrives. This assumption follows the standard distinction between naïveté and sophistication in models of present bias (O'Donoghue and Rabin, 1999). It also shapes the policy analysis. Purely endogenous commitment tools have limited bite for fully naïve users, because such users do not anticipate their future self-control problems and therefore have little reason to adopt commitment ex ante. I therefore focus on default time limits, which operate as exogenous commitment devices while preserving the option to opt out. Section 3.4 provides empirical evidence consistent with imperfect internalization of future temptation renewal, and Section 4.4 explains how the structural model allows a mass of fully sophisticated users through $\chi = 0$.

Finally, the welfare analysis treats intrinsic utility as welfare relevant and temptation as a distortion. This is a normative assumption. It follows the common long-run or ex-ante welfare interpretation in models of present bias: welfare is evaluated from the perspective that is not distorted by immediate temptation. The paper does not attempt to resolve the broader philosophical question of whether welfare should reflect the long-run planner's preferences, the short-run doer's preferences, or some choice-based criterion. Instead, I make the welfare criterion explicit and use it consistently throughout the analysis. For broader discussions of welfare analysis with nonstandard decision makers, see Bernheim and Rangel (2009) and Bernheim (2009).

[7] See Section 5.1 of DellaVigna (2018) for a discussion of timing choices in structural models of present bias.

3 Data and Institutional Background

I use the short-drama industry as the empirical setting to study how short content interacts with self-control problems. The setting has two useful features. First, short dramas are consumed sequentially in minute-long episodes, making stopping decisions frequent and tightly linked to content length. Second, the platform monetizes viewing through episode unlocking and nonlinear token top-up menus, which reveal users' planned continuation values. I first describe the industry and platform institutions, then document that shorter episodes are associated with higher completion rates across dramas. I then introduce the individual-level data and use users' unlocking and top-up choices to document misprediction, overpayment, and repeated small-package purchases consistent with self-control problems.

3.1 Institutional background

Short drama series. Short drama series combine features of serialized television and short-form video. They are typically optimized for vertical smartphone viewing and divided into many brief episodes.[8] Their defining feature is brevity: a typical series contains 40 to 100 episodes, each lasting about one minute. The plots are designed to generate immediate engagement through romance, conflict, dramatic reversals, and cliffhangers. This format makes the industry a natural setting for studying temptation and self-control problems.

Demand for short dramas is large and growing rapidly. In China, where the industry originated around early 2022, the short-drama market generated $7.0 billion in revenue in 2024, exceeding box-office sales of $5.9 billion.[9] The format has also expanded globally. For example, global short-drama platforms such as DramaBox, ReelShort, and ShortMax ranked 6th, 18th, and 33rd, respectively, in the U.S. Apple App Store Entertainment Top Charts for free apps on April 26, 2024.[10] Users are also willing to pay substantial amounts for this content. On the platform I study, the superstar drama generated $3.5 million in revenue within one month of release.

At the same time, production is low-budget and optimized for speed. Scripts are usually purchased from popular web novels and adapted into plots that prioritize immediate engagement over narrative depth. Character development and nuanced storytelling are limited, while sensational or emotionally charged themes are common. Short dramas typically feature relatively unknown actors and are directed and edited by mid-level or inexperienced professionals. Production costs are therefore low, usually below $80,000 per drama, and the full production cycle often takes only seven to 10 days.

The short-drama industry thus shares important features with platforms such as TikTok and other short-video formats: content is low-cost, mobile-optimized, rapidly consumed, and de-

[8] Because of their vertical display format, these series are sometimes called "vertical dramas."

[9] Source: *2024 China Short Drama Industry Research Report*, by Shenzhen Media Group, the Audiovisual Art Research Center of the Communication University of China, and the China Television Drama Production Industry Association.

[10] For comparison, the Apple App Store Entertainment Top Charts on the same day included related apps such as TikTok (2nd), Netflix (5th), YouTube TV (16th), and AMC Theatres (46th).

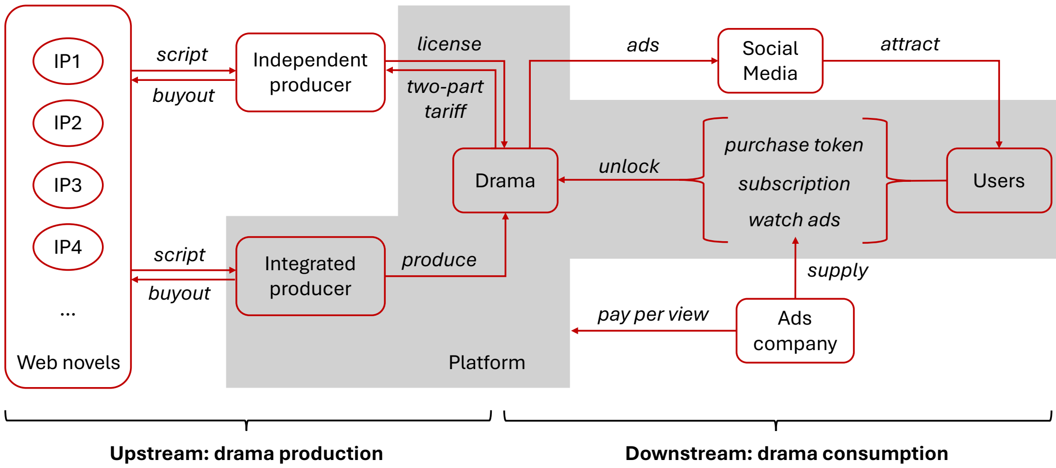


Figure 2: Institutional background of the platform

Notes: The figure summarizes the platform's core business model. Activities occurring on the platform are highlighted in the shaded area. On the supply side, scripts are acquired and adapted from web novels, then produced into short dramas either by the platform itself or by independent producers. In the latter case, the platform secures licensing rights through an upfront payment and a revenue-sharing agreement. On the demand side, the platform promotes its dramas on major social media platforms such as TikTok and Facebook to acquire users. Users unlock episodes by purchasing tokens, subscribing, or watching external ads. These activities constitute the platform's three main revenue streams. Advertising firms provide ads and pay the platform based on viewership.

signed to sustain engagement. These features make the setting well suited for studying how short content can amplify self-control problems.

The platform. The data come from a leading platform in this industry. Figure 2 summarizes the platform's business model, with activities occurring on the platform shown in the shaded area. On the supply side, scripts are purchased and adapted from web novels, then produced either by the platform itself or by external producers. When working with external producers, the platform acquires dramas through a negotiated two-part tariff consisting of an upfront payment and a revenue-sharing rule.[11] On the demand side, the platform advertises its dramas on major social media platforms such as TikTok and Facebook. Users then unlock episodes by purchasing tokens, subscribing, or watching external ads, which form the platform's three revenue streams. Appendix B provides additional institutional details.

The platform operates globally and averaged 120,000 daily active users in 2024. I focus on the U.S. market, which accounts for 23% of users and 63% of revenue. The user base is predominantly female: 86% of users are women. It is also relatively young: 36% of users are under 30, 63% are under 40, and 85% are under 50.[12] Because Facebook and TikTok are the platform's main advertising channels, its users substantially overlap with audiences on major social media platforms,

[11] The two-part tariff includes an upfront payment and a revenue-sharing rule from the platform.

[12] The platform calculated age for the period between March 1, 2024, and May 14, 2024. The app is labeled "18+," so only users aged over 18 are included. The age distribution may skew even younger because teenagers may access the platform using their parents' phones.

making them a relevant sample for studying digital addiction.

By March 2024, the platform offered more than 400 dramas across major languages. These dramas are adapted from web novels and concentrate in melodramatic genres, with the most popular categories being "love after marriage" and "toxic love."[13] On average, users consume 1.7 million episodes per day. Viewership is highly concentrated: a single superstar drama accounts for 58.8% of total viewership. The superstar drama was produced in-house at a cost of $70,000, released in mid-January 2024, and generated $3.5 million in revenue within one month. The platform also invested heavily in promotion, spending more than $600,000 on advertising during the first month. Spillovers across dramas are limited. Among users who unlocked the superstar drama, only 3.6% also unlocked another drama, and only 1.5% topped up for another drama. I therefore focus on user behavior for this superstar drama.

The platform monetizes demand through episode unlocking. Users can unlock episodes in three ways, as shown in Figure 2: by spending tokens, subscribing, or watching ads. My analysis focuses on token purchases, the platform's main revenue source. Between May 2023 and May 2024, tokens generated $31,028 in daily income and accounted for 78.0% of total revenue.[14] Because users must unlock episodes sequentially by spending a fixed number of tokens, the setting provides episode-level variation in willingness to pay. After unlocking an episode, users can rewatch it without additional payment. I focus on initial viewing and do not model repeated viewing. In the sample, users unlock episodes sequentially and typically watch each episode in full at least once before unlocking the next. This pattern supports the assumption that users make viewing decisions at the episode level.

Users purchase tokens with real money through top-up packages when their token balance is insufficient.[15] The top-up page appears automatically, and payment can be completed through the App Store or Google Play within seconds, making the purchase process low-friction. The platform offers four packages: $4.99 for tokens equivalent to 7.50 episodes, $9.99 for 16.62 episodes, $19.99 for 36.74 episodes, and $29.99 for 81.49 episodes.[16] The menu is nonlinear: larger packages have lower average prices per episode but require greater upfront commitment. This trade-off provides a source of variation in package-implied planned continuation at the purchase moment.

[13]The "love after marriage" genre explores romantic relationships that develop and evolve after marriage. "Toxic love" refers to dysfunctional relationships characterized by unhealthy behaviors such as manipulation, control, emotional abuse, and dependency.

[14]The second-largest revenue source is subscriptions. The platform offers three subscription options: one week ($29.99), one month ($59.99), and one year ($199.99). Subscriptions generated $6,631 in average daily income and accounted for 16.7% of total revenue. The final revenue source is advertising: users watch 30-second ads to unlock episodes, generating $4,329 in average daily income and 5.4% of total revenue. The time series of these revenue sources is reported in Appendix B.

[15]As part of the platform's advertising strategy, users can also earn small amounts of gift tokens through daily log-ins, sharing content on social media, and following the platform's TikTok account. These quantities are negligible relative to top-up packages.

[16]Promotions generate variation in the number of tokens included in each package across users and over time.

Table 2: Episode length and drama completion rate (%)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Episode length (minutes)	-2.18	-2.75^{**}	-3.22^{**}	-2.79^{**}	-2.58^{**}	-2.46^{***}	-2.42^{***}
	(1.52)	(1.14)	(1.28)	(1.25)	(1.02)	(0.92)	(0.92)
Fixed effects							
Language		Yes	Yes	Yes	Yes	Yes	Yes
Genre			Yes	Yes	Yes	Yes	Yes
Release-month				Yes	Yes	Yes	Yes
Controls							
Episode count					Yes	Yes	Yes
First-episode viewers						Yes	Yes
Bitrate							Yes
Observations	353	353	353	353	353	353	353
Mean dep. var.	11.44	11.44	11.44	11.44	11.44	11.44	11.44
R^2	0.00	0.73	0.74	0.77	0.78	0.80	0.80

Notes: The dependent variable is the drama-level completion rate, defined as the fraction of first-episode viewers who remain until the terminal episode. The coefficient on episode length is measured in percentage points per additional minute. Standard errors in parentheses are HC1 robust. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

3.2 Aggregate evidence on episode length and completion

Before turning to user-level choices, I use aggregate data from all drama series on the platform to document a motivating fact about content length. The outcome is the drama-level completion rate, defined as the fraction of first-episode viewers who reach the final episode of a series. The data include each drama's average episode length, number of episodes, language, release date, genre tags, video bitrate, retention rate, and viewer count.[17] I restrict the sample to dramas with positive first-episode viewership, language information, and non-missing completion rates. The resulting sample contains 353 drama series. Average episode length is 1.41 minutes, with a standard deviation of 0.28 minutes.

I estimate a sequence of drama-level reduced-form regressions:

$$\text{CompletionRate}_d = \beta \text{Length}_d + \mathbf{X}_d' \Gamma + \text{FE}_d + \varepsilon_d. \tag{7}$$

The specifications move from a length-only regression to richer controls for series characteristics. I sequentially add language fixed effects, genre fixed effects, release-month fixed effects, episode count, first-episode viewers, and video bitrate. The last three variables control for series length, initial audience size, and a proxy for technical video quality, respectively. This exercise is descriptive: episode length is chosen jointly with content, production, and marketing decisions, so the estimates should be interpreted as conditional correlations rather than causal effects.

Table 2 documents a robust negative relationship between episode length and completion.

[17] The retention data cover viewing between March 1 and March 31, 2024, and the matched dramas in the analysis were released between November 10, 2022, and March 15, 2024.

Column (1) reports the raw correlation: an additional minute of episode length is associated with a 2.18-percentage-point lower completion rate. Columns (2)–(4) add language, genre, and release-month fixed effects. The estimate becomes larger in magnitude after accounting for language and genre composition and remains negative after absorbing release-month differences across dramas. Columns (5)–(7) add episode count, first-episode viewers, and video bitrate. The coefficient remains stable across these richer specifications. In the full specification, an additional minute of episode length predicts a 2.42-percentage-point lower completion rate, with a standard error of 0.92. Relative to the mean completion rate of 11.44%, this effect equals 21.2% of the mean.

This aggregate fact motivates the paper's central mechanism. Across dramas on the same platform, shorter episodes are associated with higher completion rates. The pattern is consistent with the idea that short content lowers the stopping margin: each episode requires only a small incremental commitment, and the decision to stop is repeatedly deferred to the next short interval. The aggregate evidence does not identify a causal effect of episode length or, by itself, establish a self-control mechanism. It does, however, show that the content-length margin emphasized in the model is visible in platform-wide data. The user-level analysis below studies the behavioral mechanism behind this relationship, using top-up and unlocking choices to distinguish rational continuation from self-control problems.

3.3 Individual-level data on the superstar drama

My main analysis focuses on the superstar drama because user activity on the platform is highly concentrated on this title. The drama consists of 80 episodes. The first nine episodes are free; each subsequent episode costs 65 tokens to unlock. In the analysis below, I normalize token quantities by this per-episode price, so that one unit corresponds to the tokens required to unlock one episode. For a representative 10% random sample of global users, I observe individual-level logs covering top-ups, unlocking, and viewership from November 1, 2023, to March 31, 2024.[18] I focus on two choice margins: top-up purchases and episode unlocking. For each top-up purchase, I observe the transaction time, price paid, and number of tokens acquired. For each episode unlock, I observe the time, unlocking method, and number of tokens spent.[19] These records characterize users' token-financed consumption of the drama.

Sample construction. I summarize the data-cleaning procedure here and provide full details in Appendix C.1, including the rationale for each step, the resulting sample size, and potential selection concerns. Guided by institutional knowledge, I restrict attention to U.S. users' top-up and unlocking activities for the superstar drama. I exclude users who unlocked another drama before or during their engagement with the superstar drama, as well as users who accessed episodes through subscriptions or ad viewing. To focus on regular viewing behavior, I drop users who did not unlock episodes in their natural sequential order. To ensure that users had enough time to

[18]Each user is randomly assigned a unique user ID upon first opening the platform app. Users in my sample are those whose user ID ends with the number "8," effectively creating a 10% random sample of all users.

[19]For free episodes, I treat first-time viewership as the corresponding unlocking activity.

Table 3: Summary statistics at the user level

Variable	mean	std	min	p25	p50	p75	p90	p95	p99	max	count
Episode	19.03	23.79	1	9	9	12	80	80	80	80	11,512
Top-up ($)	5.80	13.35	0	0	0	0	29.99	39.98	49.97	69.96	11,512
Round	1.37	0.73	1	1	1	2	2	3	4	5	11,512
Gift	0.29	0.96	0	0	0	0	1	2	5	12	11,512
Endowment	0.43	0.91	0	0	0	0	2	3	4	4	11,512

Table 4: Top-up packages: Tokens and market share

Price ($)	tokens	[p5,p95]	price/episode	market share
4.99	7.50	[7,9]	0.67	0.34
9.99	16.62	[16,20]	0.60	0.36
19.99	36.74	[36,46]	0.54	0.20
29.99	81.49	[62,92]	0.37	0.11

Notes: This table reports token quantities and market shares for the four top-up packages offered by the platform. Token quantities are normalized by the per-episode token cost of 65. The column "price/episode" reports the average dollar cost of unlocking one episode under each package.

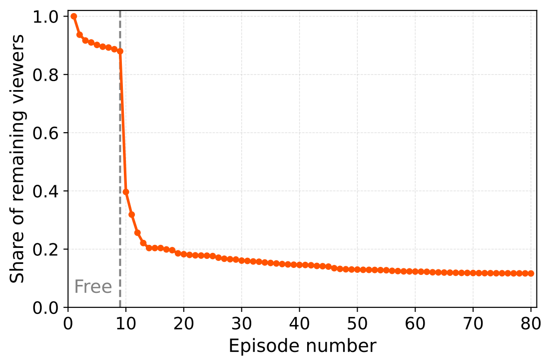
complete the drama, I retain only users who began watching before January 31, 2024, and exclude users who had not completed the series but remained active during the final week of the sample period. Finally, to ensure that tokens were primarily acquired through top-up purchases, I exclude users whose initial or gift-token endowment exceeded the equivalent of five episodes.[20]

For the resulting sample, I define a viewing "round" as a sequence of episode unlocks in which any two consecutive unlocks occur within two hours. This definition captures users who pause and later resume viewing.[21] I further exclude users in the top percentile of the round-count distribution, users who purchased top-up packages not listed in Table 4, and users whose packages contained unusual token quantities with frequency below 5% conditional on price. The final sample contains 219,064 unlocking events and 5,360 top-up purchases by 11,512 users.

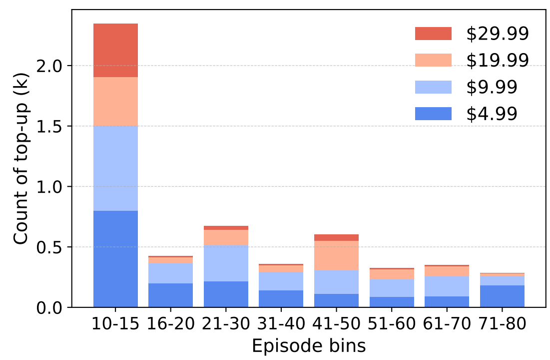
Summary statistics. Table 3 reports summary statistics. On average, a user unlocks 19.03 episodes and spends $5.80 on the drama. User behavior is highly heterogeneous. More than three-quarters of users do not purchase any tokens, and the median user stops at episode 9, the end of the free trial. More than half of users consume the drama in a single round. Gift tokens and initial endow-

[20] Users could obtain initial or gift tokens through platform promotions such as daily log-ins or sharing the drama on social media. I exclude users whose non-top-up token value exceeds five episodes, which accounts for 8.7% of the sample. This small share suggests that most users did not rely heavily on such tokens, and that the exclusion assumption is unlikely to affect the quantitative results. A full discussion, including the distribution of initial and gift tokens, is provided in Appendix C.1.

[21] I model round formation as a stochastic and exogenous process in the structural model. This simplifying assumption abstracts from pausing behavior, which is not central to the analysis. Section 4.4 discusses round dynamics in greater detail, including the rationale, robustness checks, and potential biases. All results are robust to alternative round definitions, including a 12-hour threshold.



(a) Share of remaining viewers by episode



(b) Package purchases by episode bins

Figure 3: Dynamic patterns in unlocking and top-up choices

Notes: Panel (a) reports the share of users who remain at each episode. The measure is mechanically nonincreasing because users must unlock episodes sequentially. The sample contains 11,512 users. The first nine episodes are free, which accounts for the sharp drop at episode 10. Panel (b) reports top-up package purchases by episode bins.

ments are small: on average, they correspond to 0.29 and 0.43 episodes per user, respectively.[22]

Table 4 summarizes top-up choices. The platform uses a nonlinear pricing menu: larger packages require higher upfront spending but offer a lower average price per episode. This menu creates a trade-off between lower marginal cost and greater purchase-time commitment. All four packages have positive market shares, which is important for identification. Observed package choices provide information about planned continuation, while variation across package sizes helps discipline the distribution of temptation horizons. In addition, promotional variation in token quantities across users and over time helps identify price sensitivity.

Figure 3a shows the share of users who remain at each episode. By construction, all 11,512 users in the sample watch the first episode. Viewership falls after each episode, with the largest drop at episode 10, immediately after the nine-episode free trial ends and users must purchase tokens to continue.[23] Three-quarters of users exit without purchasing tokens, while 1,340 users, or 11.64%, complete the entire drama.

Figure 3b shows the dynamic pattern of top-up choices by episode bin. The largest package, priced at $29.99, is purchased primarily between episodes 10 and 15 by highly engaged users willing to commit to many future episodes. The other three packages have positive sales throughout the series. The persistent market share of lower-priced packages across later episodes is consistent with short package-implied continuation followed by additional viewing and repurchase. Section 3.4 examines this pattern directly.

[22]Users can receive free tokens as rewards for platform activities such as daily log-ins or sharing content on social media. The quantity of these gift tokens is small, making such promotions negligible for the analysis. For simplicity, I model these tokens as additional tokens included with top-up packages in the quantitative analysis.

[23]The probability of stopping peaks at episode 10 but remains elevated in subsequent episodes, because some users have small token endowments, sufficient for one to four additional episodes, and delay their top-up decision until exhausting their token balance.

3.4 Evidence of self-control problems

This section documents behavioral patterns that are consistent with self-control problems under the maintained package-choice interpretation and that provide variation for disciplining temptation in Section 4. The key source of variation is the nonlinear pricing of top-up packages. Figure 4a illustrates the idea with an actual user who repeatedly purchases the smallest top-up package and completes the entire drama in a single night. The $4.99 package implies limited planned continuation relative to larger packages with lower per-episode prices. The user's subsequent continuation is therefore consistent with a failure to follow the earlier package-implied plan. The remainder of the section shows that this pattern is widespread, persistent, and difficult to explain with learning or other rational motives alone.

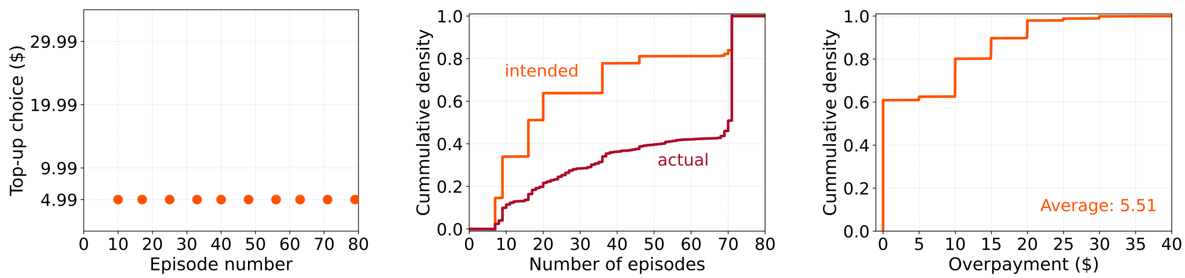
Figure 4b compares package-implied planned viewing with realized viewing among users who purchased tokens. The package-implied distribution, shown by the orange cumulative density curve, is inferred from users' first token package choice. On average, the first top-up implies planned viewing of 28.0 episodes. By contrast, realized viewing, shown by the red curve, first-order stochastically dominates the package-implied distribution. Users ultimately watch 51.0 episodes on average, 82.1% more than implied by their first package choice. This discrepancy is consistent with self-control problems under the package-choice interpretation.

The nonlinear pricing menu also yields a payment-based measure of the same inconsistency. Figure 4c reports overpayment relative to the ex-post cost-minimizing package sequence among paying users.[24] In total, 39.1% of these users spend more than the ex-post cost-minimizing benchmark for the episodes they ultimately unlock. Specifically, 17.6% overpay by approximately $10, 9.5% by $15, and 8.2% by $20. Average overpayment is $5.51, or 22.7% of optimal expenditure. This payment-based evidence complements the package-implied viewing gap, while leaving room for rational motives that the structural model addresses explicitly.

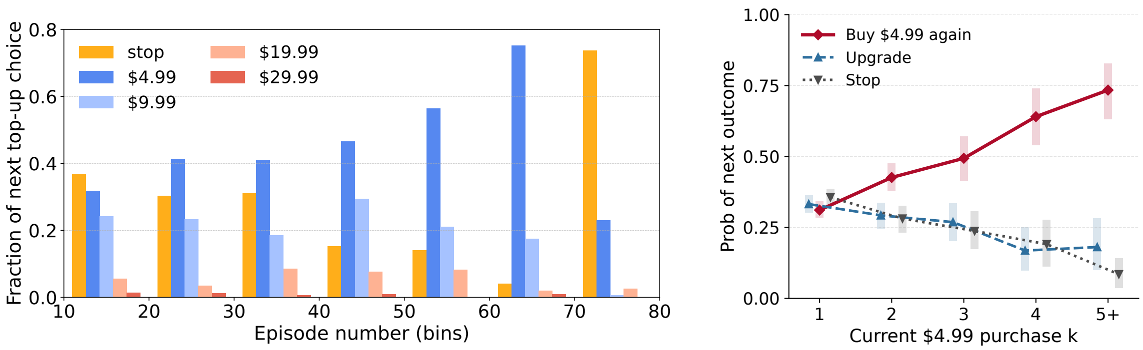
Other rational motives. The preceding evidence does not, by itself, rule out rational motives for small-package purchases. Learning is the most natural alternative explanation. A user who is uncertain about drama quality may initially buy the smallest package to acquire information, making a $4.99 purchase a plausible trial choice. As experience accumulates, however, this trial motive should weaken. Favorable signals should make larger packages more attractive, while unfavorable signals should lead users to stop. This implication is sharpened by the martingale property of Bayesian beliefs: a user's current belief is her best forecast of her future belief, so the next signal is not expected, ex ante, to move her valuation systematically in one direction, let alone toward repeated small trials. Learning should therefore both sort users across actions and reduce the incentive to keep experimenting with the smallest package.

Panels 4d and 4e show the opposite pattern. Panel 4d shows that, after purchasing the $4.99 package, many users continue buying it; moreover, this tendency increases as they watch more

[24]Optimal spending is imputed using the average token quantity in each package from Table 4.



(a) Self-control problem example (b) Intended vs. realized usage (c) Distribution of overpayment



(d) Next top-up package choice following $4.99 purchase (e) Repeated $4.99 purchases

Figure 4: Evidence of self-control problems

Notes: Panel (a) illustrates a user who repeatedly purchases the smallest top-up package. The y-axis lists the available top-up options, and the x-axis indicates the episode before which each top-up occurs. Panel (b) compares the cumulative distributions of package-implied and realized episodes watched among users who purchased token packages. Package-implied planned viewing is inferred from the first package purchased under the maintained package-choice interpretation. Panel (c) presents the cumulative distribution of overpayment among users who made at least one package purchase. Overpayment is measured relative to optimal spending, imputed using the average token quantity in each package from Table 4; its natural lower bound is $0. Panel (d) reports users' next actions after a $4.99 purchase, separately by episode group. The panel uses accumulated viewing experience as a proxy for how much uncertainty about drama quality has been resolved. Under a learning-only explanation with stationary drama quality, stable outside options, and negligible repurchase frictions, users who have watched more episodes should be less likely to keep experimenting with the same small package and more likely to upgrade or stop as beliefs move upward or downward. Instead, many users continue purchasing the $4.99 package even after substantial viewing experience. The sharp increase in stopping between episodes 70 and 80 is mechanical, because users must stop after completing the series. Panel (e) sharpens the same test by conditioning on the order of repeated $4.99 purchases. It reports the next outcome after the kth $4.99 purchase. Rectangular bars denote 95% confidence intervals from a user-level bootstrap. The probability of another $4.99 purchase rises with k, while the probabilities of upgrading and stopping fall. This persistence in repeated small purchases provides the empirical variation used to separate temptation from learning.

episodes and accumulate more information.[25] Panel 4e focuses directly on this repeated-purchase margin by asking what users do after their kth \$4.99 purchase. The probability of buying the \$4.99 package again rises from 31.1% after the first such purchase to 73.4% after at least five such purchases. After the fifth or later \$4.99 purchase, only 18.1% of users upgrade and 8.5% stop. This increasing persistence in small-package repurchases is difficult to reconcile with a learning-only account in which experience resolves uncertainty; instead, it points to a persistent local-repurchase force consistent with self-control problems.

Appendix C.4 complements the transition evidence with a conservative purchase-time package-replacement exercise, following the payment logic in DellaVigna and Malmendier (2006). For each episode bin, I replace the user's first \$4.99 purchase with a larger package available in the same menu, hold realized viewing fixed, and do not allow later top-up choices to be reoptimized. The exercise is conservative because it leaves room for rational motives for choosing the smallest package, such as learning, outside-option uncertainty, and unused-token concerns; only positive net savings are interpreted as payment evidence of overpayment. Despite this conservative accounting, late \$4.99 purchases imply positive savings. Replacing the first \$4.99 package with the \$9.99 package saves \$0.20, \$0.30, and \$2.62 per user on average in the 40–50, 50–60, and 60–70 episode bins, respectively. These positive late-bin entries provide lower-bound evidence that repeated small-package purchases become costly after substantial viewing experience.

The structural model in Section 4 therefore allows users to learn about drama quality, but it also targets the conditional transition probabilities in Panels 4d and 4e as moments that help disentangle learning from temptation. Learning is disciplined by the extent to which users sort into upgrading or stopping, whereas temptation is disciplined by the persistent and increasing tendency to keep purchasing the smallest package.

Sophistication. The repeated small-package evidence is also informative about sophistication. The top-up menu is not an effective commitment device: after a user exhausts a small package, she can immediately buy tokens again. A fully sophisticated but still tempted user would understand this lack of commitment. If she expects future selves to keep topping up, she may prefer a larger package upfront to lower the expected per-episode price, rather than repeatedly buying the smallest package. Thus, the data do not rule out some sophisticated users, but the persistent return to the smallest package after substantial viewing experience is difficult to reconcile with a model in which all tempted users fully anticipate and internalize the future renewal of temptation.

Underlying mechanisms. This section presents evidence of self-control problems while remaining agnostic about the precise behavioral mechanism underlying the observed time inconsistency. In the model, short-run temptation can be interpreted as present bias that overweights immediate entertainment, inertia that makes stopping difficult, or another local force that raises the perceived value of continuation relative to welfare-relevant utility. The data do not contain the variation

[25] The spike in stopping behavior between episodes 70 and 80 is mechanically high, because users must stop after completing the series.

needed to distinguish among these foundations. For the empirical margins studied here, they operate through the same wedge between perceived short-run value and welfare-relevant continuation value. I therefore use temptation as a reduced-form framework for analyzing the problem.

4 Structural model

I adopt a structural approach to systematically examine self-control problems in the short-drama-series industry while allowing users to learn about drama quality. Building on the stylized model from section 2, I develop a single-agent dynamic discrete-choice model in section 4.1. The model solution is characterized in section 4.2. Section 4.3 reports main predictions on self-control problems, followed by a discussion of modeling assumptions and interpretation in section 4.4.

4.1 Setup

A superstar drama consists of T episodes indexed by t. Each one-minute episode defines a decision period in which the user decides whether to purchase a top-up package and unlock the next episode. The drama-watching activity can therefore be modeled as a finite-horizon dynamic discrete-choice problem. This section presents the single-agent setup; I omit the individual subscript i for simplicity.

Welfare-relevant utility. Let the true quality of the drama for this user be fixed at δ per episode, where δ varies across users according to $\mathcal{N}(\mu_\delta, \sigma_\delta^2)$. I define the welfare-relevant utility from consuming episode t as

$$u_t^*(h_t, \delta, \psi_t, \epsilon_t) := \underbrace{\overbrace{\delta}^{\text{quality}} - \overbrace{\psi_t}^{\text{outside}}}_{\text{intrinsic utility}} + \underbrace{\alpha(h_t)}_{\text{habit formation}} + \underbrace{\epsilon_t}_{\text{taste shifter}}, \tag{8}$$

which consists of intrinsic utility, habit formation, and a taste shifter. The intrinsic utility captures the value of the drama relative to the user's outside option ψ_t. The taste shifter ϵ_t captures episode-specific shocks to the user's utility and is unknown to the user before time t.

I allow for rational addiction through habit formation (Becker and Murphy, 1988).[26] When encountering episode t, the user has habit stock h_t, defined as the number of previous episodes watched within the current round r_t. This stock depreciates across rounds. Habit stock enters utility through the nonlinear function $\alpha(h_t) = \alpha_1 h_t (h_t - \alpha_2)$.[27] Rounds evolve exogenously: after

[26] In drama series, episodes are linked by a continuous storyline, which helps keep users engaged and makes habit formation relevant. Appendix C.2 provides evidence on habit formation by examining the relationship among habit stock, the probability of continuing watching, and the probability of purchasing higher-priced packages.

[27] I adopt this quadratic functional form for the following reasons. First, it satisfies $\alpha(0) = 0$, implying no additional utility in the absence of habit stock. Second, when $\alpha_1 < 0$, it generates an inverted-U shape, capturing diminishing returns due to fatigue. Third, the parameters are straightforward to interpret. This functional form is not critical to my

watching an episode, the user must start a new round with probability ρ, so that $r_{t+1} = r_t + 1$.[28] By design, habit formation is temporary: it accumulates within a round and captures the transient nature of digital addiction. I assume that users have rational expectations about both habit formation and round dynamics, and fully internalize these forces when making decisions.

In practice, I assume that the outside value ψ_t is round-specific and depends on the time of day at which the user begins watching. I normalize $\psi_t = 0$ for rounds that start during the day, defined as 11am–12am, and parameterize the outside value for nighttime, defined as 1am–10am, as $\overline{\psi}$.[29] The probability that a new round begins at night, $\varphi = 0.27$, is calibrated directly from the data. I also assume that the random taste shifter ϵ_t follows a Gumbel distribution, $Gumbel(-\beta_\epsilon \gamma, \beta_\epsilon)$.[30]

Perceived flow utility. When making choices, however, the user perceives flow utility as:

$$u_t(h_t, \widehat{\delta}_t, \psi_t, \chi, \epsilon_t) = \underbrace{\overbrace{\widehat{\delta}_t}^{\text{belief}} - \overbrace{\psi_t}^{\text{outside}}}_{\text{expected intrinsic utility}} + \underbrace{\mathbb{1}\{\chi \geq 1\}\kappa}_{\text{temptation}} + \underbrace{\alpha(h_t)}_{\text{habit formation}} + \underbrace{\epsilon_t}_{\text{taste shifter}} . \tag{9}$$

The perceived utility differs from welfare-relevant utility in two respects: it includes a short-run, irrational temptation term, and it replaces true drama quality with the user's belief, which evolves through rational learning.

First, following the stylized model in Section 2, I capture self-control problems as short-run temptation with naïve perception. This formulation can be microfounded by behavioral mechanisms such as present-biased preferences or inertia.[31] The per-minute temptation κ is active when its duration χ exceeds the one-minute episode length.[32]

Second, the user does not directly observe the fixed drama quality δ. Instead, before episode t, the user holds a posterior belief with mean $\widehat{\delta}_t$. Users enter the drama with a rational-expectations prior. After watching episode t, the user observes a clean signal $x_t = \delta + \eta_t$, where $\eta_t \sim \mathcal{N}(0, \sigma_\eta^2)$ captures episode-level experience noise and is distinct from the taste shifter ϵ_t. By Kalman filter,

quantitative results.

[28] The assumption of exogenous round dynamics is crucial for identifying habit formation. It is reasonable because if users stop watching because they dislike the series, they should not resume in the next round after losing all accumulated habit stock. Thus, any observed temporary disruptions must stem from exogenous factors—for example, a user might receive an unexpected email about a paper decision, interrupting their drama-watching session. See section 4.4 for a complete discussion of this round dynamic.

[29] This choice of time window is motivated by viewing patterns in the data, which show below-average viewership between 1am and 10am. A detailed description can be found in Appendix C.3.

[30] The parameter β_ϵ is the scale parameter that controls the dispersion of the Gumbel distribution. The location parameter is set to $-\beta_\epsilon \gamma$, where γ is the Euler constant, so that the expectation of ϵ_t is zero (e.g., Rust, 1987).

[31] I remain agnostic about the precise mechanism. These mechanisms are modeling-wise isomorphic, because different interpretations of the temptation term yield the same welfare implications.

[32] For simplicity, I assume that temptation remains constant every minute within its duration, although a more flexible model could allow it to decay over time. Although I cannot estimate the rate at which temptation decays, incorporating this feature would only strengthen the present bias, further amplifying the self-control problem and making the short video length an even greater factor in driving addiction.

the posterior before episode t is $\mathcal{N}(\widehat{\delta}_t, \widehat{\sigma}_t^2)$, where

$$\widehat{\sigma}_t^2 = \left(\frac{t-1}{\sigma_\eta^2} + \frac{1}{\sigma_\delta^2} \right)^{-1} \quad \text{and} \quad \widehat{\delta}_t = \widehat{\delta}_{t-1} + \frac{\widehat{\sigma}_{t-1}^2}{\widehat{\sigma}_{t-1}^2 + \sigma_\eta^2} \left(x_{t-1} - \widehat{\delta}_{t-1} \right). \tag{10}$$

Because the posterior variance is deterministic in the number of watched episodes, the stochastic learning state is summarized by $\widehat{\delta}_t$.

Dynamics. I now formalize the dynamic components of the model. To match the data, I condition on users having watched the first episode and initialize their beliefs at the prior mean, $\widehat{\delta}_1 = \mu_\delta$. Each user has a type-specific effective temptation horizon $\chi \in \{0, 1, \ldots, 80\}$, measured in minutes. The type $\chi = 0$ captures users whose choices are not distorted by the rolling-temptation channel. This group includes users who experience no temptation and may also include fully sophisticated users who anticipate and neutralize future temptation. The model therefore does not require all users in the sample to be naïve.

Conditional on positive effective temptation, $\chi > 0$, users have naïve beliefs about the future renewal of temptation. At any decision point, the current self expects the current temptation horizon to run down. In realized behavior, however, each future self again faces the same type-specific horizon χ. Temptation therefore rolls forward over the viewing sequence. The upper bound of 80 minutes equals the length of the focal drama; it preserves the interpretation of temptation as short-run relative to the full series and improves computational tractability.

At the beginning of each episode $t \geq 2$, the user has unlocked and watched episodes $1, \ldots, t-1$. The platform sets the normalized token cost to $c_t = \mathbb{1}\{t > 9\}$, so the first nine episodes are free and each subsequent episode costs one normalized token. The user's state is $\mathbf{s}_t = (h_t, a_t, \widehat{\delta}_t)$, where h_t is habit stock, a_t is token balance, and $\widehat{\delta}_t$ is the posterior mean of drama quality.[33] Figure 5 summarizes the timing. In each period, the user first decides whether to purchase a top-up package if her token balance is insufficient, and then decides whether to unlock the next episode.

First, the user decides whether to top up when the current token stock a_t is below the cost of unlocking the next episode, c_t.[34] I assume that the top-up process is frictionless, because users can complete purchases within seconds.[35] They choose from a menu of $K = 4$ packages, indexed by $k \in \{1, \ldots, K\}$, with price p_k and token quantity b_{kt}, as well as an outside option denoted by $p_0 = b_0 = 0$. Packages are ordered so that p_k increases with k. Token quantities are randomly drawn for each user-period pair from a platform-set distribution $F_{\boldsymbol{b}}(\cdot)$, reflecting random promotions. In practice, the platform uses nonlinear pricing: the average price per token, $\mathbb{E}[p_k / b_{kt}]$,

[33] The posterior variance $\widehat{\sigma}_t^2$ is pinned down by t and therefore does not need to be carried as a separate stochastic state.

[34] In practice, the top-up page only appears when users have insufficient token balance. Users could, in principle, stop watching, return to the main page, and manually access the top-up page at any time, but fewer than 1% of users choose to do so when they have sufficient tokens.

[35] This assumption is quantitatively conservative because introducing friction would reduce users' willingness to top up, implying that the structural model would require a larger temptation component to account for the observed repeated top-up behavior.

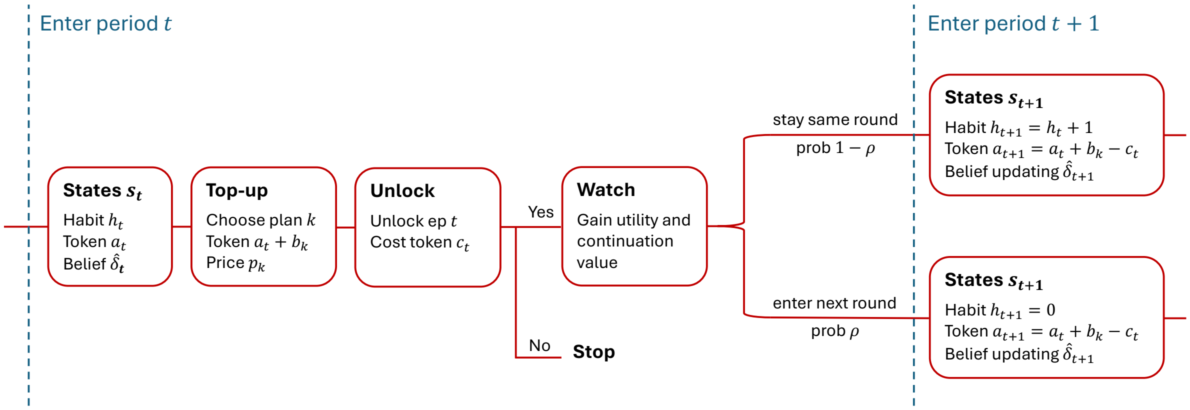


Figure 5: Timing: Decisions on top-up and unlocking within a time period

decreases with package price. This pricing structure creates a trade-off between lower per-token prices and greater commitment to future viewing. Top-up choices therefore provide a package-implied measure of planned continuation at the purchase moment, given the user's current beliefs, temptation state, token balance, and package-specific taste shock. I also assume an idiosyncratic taste shifter ν_{kt} for each top-up package, distributed as standard Gumbel. I denote the value and expected value functions associated with the top-up decision by $V_t(\mathbf{s}; \psi, \chi, \boldsymbol{b}, \boldsymbol{\nu})$ and $EV_t(\mathbf{s}; \psi, \chi)$, respectively, where state variables appear before the semicolon.

Second, after any top-up decision, the user decides whether to unlock episode t, given habit stock h_t, token stock $a_t + b_{k^*t}$, and belief $\widehat{\delta}_t$. If the user unlocks the episode, they expect to receive perceived flow utility u_t, defined in equation (9), together with a perceived continuation value, denoted by $\mathcal{C}_t(\mathbf{s}; \psi, \chi)$ for notational simplicity. This continuation value comes from the option to unlock episode $t+1$. When evaluating this option, users perceive themselves to be one minute less tempted under the naïveté assumption and form Bayesian expectations over the next belief $\widehat{\delta}'$. I denote the perceived value and expected value functions associated with the unlocking decision by $W_t\,(\mathbf{s}; \psi, \chi, \boldsymbol{\epsilon})$ and $EW_t\,(\mathbf{s}; \psi, \chi)$, respectively.

Finally, I assume that neither habits nor tokens generate continuation value outside this superstar drama. This yields the terminal condition

$$EV_{T+1}(\cdot) \equiv EW_{T+1}(\cdot) \equiv 0. \tag{11}$$

This assumption is motivated by the empirical observation that spillovers across dramas are minimal and that most users do not watch other content after completing this drama.[36] The terminal condition in (11) therefore makes the user's dynamic problem finite horizon.

[36] See more details in Appendix C.5.

4.2 Solution: Recursive definition of value functions

Following O'Donoghue and Rabin (1999), I use the concept of *perception-perfect equilibrium strategies* to account for the behavioral forces in my model. Under this framework, users maximize their *perceived* utility, which can lead to irrational self-control problems. I define the value functions iteratively by using backward induction, which, combined with the terminal condition (11), fully characterizes the solution.

For any episode $t \leq T$, the perceived value function associated with the top-up decision for a user with habit stock h, token stock a, posterior mean $\widehat{\delta}$, outside value ψ, token amounts b_k, perceived temptation duration χ, and taste shifter ν_k is given by:

$$\begin{aligned} V_t(\mathbf{s};\psi,\chi,\boldsymbol{b},\boldsymbol{\nu}) = \quad & \mathbb{1}\{a < c_t\} \max_{k\in\{0,1,\ldots,K\}} \left\{ EW_t(h, a+b_k, \widehat{\delta}; \psi, \chi) - \omega p_k + \nu_k \right\} \\ & + \mathbb{1}\{a \geq c_t\} EW_t(h, a, \widehat{\delta}; \psi, \chi), \end{aligned} \tag{12}$$

where $EW_t(\cdot)$ is the perceived value from unlocking and ω represents price sensitivity. If the token balance is insufficient ($a < c_t$), the user must choose one of the top-up options k; otherwise, they skip topping up and proceed to the unlocking stage. Because the taste shifter ν_{kt} follows a standard Gumbel distribution, the expected value can be written as:

$$\begin{aligned} EV_t(\mathbf{s};\psi,\chi) = \quad & \mathbb{1}\{a < c_t\} \mathbb{E}_{\boldsymbol{b}} \left[\log \left(\sum_{k=0}^{K} \exp \left[EW_t(h, a+b_k, \widehat{\delta}; \psi, \chi) - \omega p_k \right] \right) \right] \\ & + \mathbb{1}\{a \geq c_t\} EW_t(h, a, \widehat{\delta}; \psi, \chi), \end{aligned} \tag{13}$$

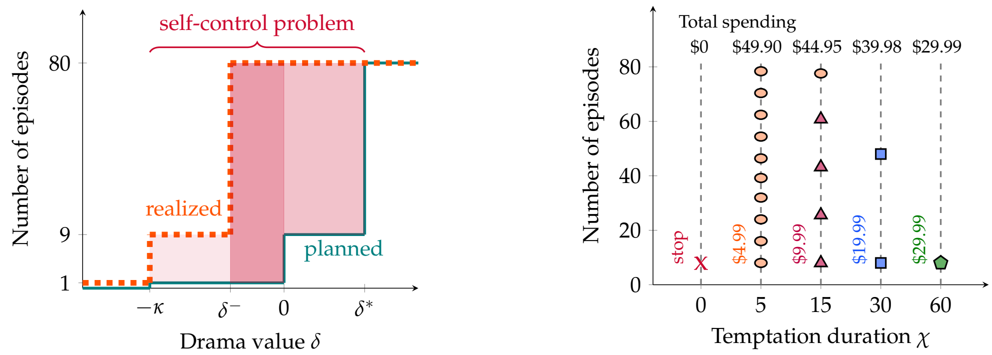
where the expectation $\mathbb{E}_{\boldsymbol{b}}(\cdot)$ is taken over the token amount $\boldsymbol{b}$.

In the unlocking stage, the user decides whether to unlock episode t. By unlocking, they derive the perceived flow utility $u_t(h, \widehat{\delta}, \psi, \chi, \epsilon)$ and gain the perceived continuation value defined as:

$$\begin{aligned} \mathcal{C}_t\left(\mathbf{s};\psi,\chi\right) = \quad & (1-\rho)\, \mathbb{E}_{\widehat{\delta}'|\widehat{\delta}} \left[EV_{t+1}\left(h+1, a-c_t, \widehat{\delta}'; \psi, \max\left\{\chi-1, 0\right\}\right) \right] \\ & + \rho \mathbb{E}_{\psi', \widehat{\delta}'|\widehat{\delta}} \left[EV_{t+1}\left(0, a-c_t, \widehat{\delta}'; \psi', \max\left\{\chi-1, 0\right\}\right) \right], \end{aligned} \tag{14}$$

where the user forms rational expectations over the future habit stock, outside value, and belief update. The next posterior mean $\widehat{\delta}'$ follows equation (10). In the current self's perceived continuation problem, naïveté makes the remaining temptation horizon decline by one minute when evaluating episode $t+1$, producing $\max\{\chi-1,0\}$ in (14). In realized choices, however, the future self again evaluates continuation with the type-specific horizon described above. The value function associated with this unlocking stage is therefore:

$$W_t\left(\mathbf{s};\psi,\chi,\boldsymbol{\epsilon}\right) = \mathbb{1}\{a \geq c_t\} \max\left\{ u_t(h,\widehat{\delta},\psi,\chi,\epsilon) + \mathcal{C}_t(h,a,\widehat{\delta};\psi,\chi), \epsilon_0 \right\} + \mathbb{1}\{a < c_t\}\epsilon_0, \tag{15}$$



(a) Medium-valued users overconsume most

(b) Longer-tempted users overpay less

Figure 6: Illustrative examples: Self-control problems as functions of δ and χ

Notes: This figure illustrates self-control problems under different values of drama quality δ and temptation horizon χ. To isolate the relevant mechanisms, I assume that drama quality δ is known to consumers, set all taste shocks (ϵ and ν) to zero, and assume no initial token endowment. Panel (a) plots the current-plan number of episodes watched (teal, solid line) and the realized number of episodes watched (orange, dashed line) for a tempted user with a one-minute temptation horizon ($\chi = 1$), as functions of δ. The shaded red area represents the magnitude of the self-control problem, measured as the gap between current-plan and realized consumption; darker shading indicates a more severe problem. Panel (b) illustrates the sequence of top-up choices for a user who, absent temptation, would watch only the free episodes but, under temptation, completes the entire series. The panel varies temptation horizon χ and reports the corresponding total spending, showing that longer temptation horizons are associated with lower overpayment.

which, with $\epsilon \sim Gumbel(-\beta_\epsilon \gamma, \beta_\epsilon)$, yields the expected value function:

$$EW_t(\mathbf{s}; \psi, \chi) = \mathbb{1}\{a \geq c_t\} \beta_\epsilon \log\left(1 + \exp\left(\frac{\widehat{\delta} - \psi + \mathbb{1}\{\chi \geq 1\}\kappa + \alpha(h) + \mathcal{C}_t(\mathbf{s}; \psi, \chi)}{\beta_\epsilon}\right)\right). \quad (16)$$

The solution of this single-agent dynamic model is characterized by the recursive system of equations (12)–(16), the belief transition (10), and the terminal condition (11). Numerically, one can solve these value functions and the corresponding policy rules by iterating over posterior mean $\widehat{\delta}$, temptation duration χ, and episode t. I further characterize model solutions in Appendix D.1. There, I analyze users' top-up and unlocking choices, establish the cutoff values of posterior quality that govern package selection, and show how variation in temptation duration shapes demand for different package sizes. I also demonstrate that the unlocking decision follows a threshold rule in posterior mean, habit stock, and temptation.

4.3 Predictions of Self-Control Problems

This section characterizes how self-control problems vary with drama value δ and temptation duration χ. These two dimensions generate distinct behavioral margins: overconsumption in the number of episodes watched and variation in top-up package choices. To isolate these mechanisms, I abstract from learning by imposing perfect information, set taste shocks ϵ and ν to zero, assume no initial token endowment, and fix each top-up package at the average token quantity.

Figure 6a illustrates an inverted-U relationship between drama value δ and the magnitude of the self-control problem, measured by the shaded gap between current-plan and realized episodes watched. Users with low valuations ($\delta < -\kappa$) do not continue beyond the first episode, even when tempted, and therefore do not suffer from a self-control problem. Users with high valuations ($\delta > \delta^*$) rationally choose to watch all episodes, so temptation does not distort their behavior. Self-control problems instead arise primarily among users with intermediate valuations ($-\kappa < \delta < \delta^*$), for whom temptation drives a wedge between the current plan and realized consumption. In the example with a one-minute temptation horizon ($\chi = 1$), users with $-\kappa < \delta < 0$ plan to watch only the first episode, while users with $0 < \delta < \delta^*$ plan to watch only the free episodes, as shown by the teal line. Temptation shifts realized consumption upward, as shown by the orange line. Users with $-\kappa < \delta < \delta^-$ watch nine episodes rather than the one episode in the current plan. Users with $\delta^- < \delta < \delta^*$ watch the entire series, even though their current plan covers only one episode when $\delta^- < \delta < 0$ and only the free episodes when $0 < \delta < \delta^*$. This pattern has direct welfare implications under the maintained long-run welfare criterion: temptation-driven self-control problems are most severe for users with mid-range valuations. The structural estimates in Section 6.1 confirm this result.

Figure 6b shows how the temptation horizon χ affects top-up choices and overpayment. Without temptation ($\chi = 0$), the user watches only the free episodes. With temptation ($\chi > 0$), the user instead pays to finish the entire series in this illustrative example. The package-implied number of episodes at the time of purchase varies with χ, so users with longer temptation horizons are more likely to buy larger token packages. As a result, although all tempted users in the example consume the full series, overpayment declines with χ: higher-χ users choose contracts with lower per-episode prices. This variation, generated by the menu of nonlinear contracts, helps discipline the distribution of temptation duration and provides a novel contribution to work using contract choices to study self-control (e.g., DellaVigna and Malmendier, 2006).

Together, these examples clarify the behavioral margins that identify temptation in the structural model. Variation in drama value δ governs the gap between current-plan and realized consumption, and therefore disciplines the welfare-relevant incidence of self-control problems. Variation in temptation duration χ governs package choice. Users with short temptation horizons plan only limited continuation and therefore tend to choose smaller packages, while users with longer temptation horizons anticipate more episodes and are more likely to choose larger packages with lower per-episode prices. The market shares of the four top-up packages therefore provide information about the distribution of temptation durations, not only the average strength of temptation. Repeated purchases of small packages discipline the magnitude of local temptation, whereas the relative shares of small, medium, and large packages discipline how long temptation persists across users. This distinction is central to the paper's contribution relative to contract-choice evidence on self-control problems: the rich top-up menu helps recover the distribution of temptation horizons. I build the estimation strategy around these margins in Section 5.1.

4.4 Discussion

I discuss my model assumptions in this section to make clear which features are identified from the data, which are maintained behavioral assumptions, and how those assumptions may affect the interpretation of the estimates.

Sophistication. The model does not require every user to be naïve. The mass at $\chi = 0$ is interpreted as a behaviorally undistorted type with respect to rolling temptation. This group includes users who experience no temptation and may also include users who anticipate and effectively neutralize future temptation. The maintained naïveté assumption applies only to users with positive effective rolling temptation, $\chi > 0$: when forecasting future choices, they do not fully internalize that temptation will renew at later stopping decisions. This interpretation nests a limited form of sophistication. The data do not separately identify users who face no temptation from users who are tempted but sufficiently sophisticated to behave as if their choices were undistorted. Accordingly, the estimated mass at $\chi = 0$ should be read as the share of users with zero effective rolling temptation, not necessarily as the share with no psychological temptation.

This nesting result is deliberately narrow. It does not cover partially sophisticated users who anticipate future temptation but remain behaviorally affected by it. Because the top-up menu is not a commitment device, such users would not view a small package as binding future consumption: after depleting tokens, they can always top up again. If they expect to continue, they may instead choose larger packages upfront to reduce expected per-episode costs. The repeated small-package behavior in the data is therefore difficult to reconcile with a model in which all tempted users are fully sophisticated. The baseline model instead captures the behaviorally relevant component of naïveté among users with positive effective temptation, while allowing the zero-effective-temptation type to absorb undistorted users, including sophisticated users who neutralize temptation.

Naïveté about habit formation. I model irrationality in my framework through temptation and the associated naïveté. As summarized in Allcott, Gentzkow, and Song (2022), irrational digital addiction can also stem from naïveté about habit formation. Although this type of irrationality could also lead to repeated top-up purchases, we would expect users to gradually shift to higher-priced top-up plans as their habit stock grows over time. However, this pattern is inconsistent with what is shown in Figure 4d, where users frequently purchase another $4.99 package, regardless of how many episodes they have watched. Thus, I conclude naïve temptation is the more relevant story in the context of this short drama.

Round dynamics. I model round dynamics as exogenous, capturing users who pause and later resume viewing—potentially due to a temporary increase in the outside option. This assumption provides a tractable way to account for pausing behavior, which is not the focus of the analysis. It is not critical for the quantitative results, because all findings remain robust when rounds are defined using alternative thresholds, such as 12 hours. Alternatively, one could model pausing endogenously by assuming decreasing returns to viewing due to fatigue. My non-linear habit formation specification $\alpha(h)$ allows for this mechanism, whereas the estimation results in Section 5.2 reject decreasing returns.

Constant episode value. Instead of allowing the underlying drama quality to vary systematically by episode, I assume users learn about a constant latent value δ. This simplification is supported by the observation that viewership evolves smoothly across episodes in Figure 3a. If episode values differed systematically, we would expect greater volatility in viewership patterns. In the learning specification, η_t captures mean-zero episode-level experience noise used for belief updating, while the Gumbel shifter ϵ_t captures idiosyncratic taste shocks at the unlocking margin.

Network externality. Social media platforms often benefit from network externalities, which can contribute to user addiction (Barwick, Chen, Fu, and Li, 2024) or heighten the fear of "missing out" (Bursztyn, Handel, Jimenez, and Roth, 2023). I exclude this mechanism from my structural model for two main reasons. First, the platform analyzed here is small relative to mainstream social media, limiting potential network effects. Second, user utility on this platform is primarily driven by content–the short dramas–rather than social interaction, because users cannot message each other or leave comments on episodes.

Pacing. An alternative explanation for the repeated purchase of small top-up packages is that users are pacing themselves for reasons such as liquidity constraints or mental budgets. This explanation is unlikely to account for the full pattern in my context. First, even the most expensive package ($29.99) is affordable for most adults. Second, as shown in Table 3, over half of users complete their viewership within a single round (a two-hour window), and more than 90% do so within two rounds. The short time frame makes repeated budgeting less likely to explain all persistent small-package repurchases.

5 Estimation

I estimate my demand model using individual-level data from the short drama platform. The estimation approach is detailed in section 5.1, and the results are reported in section 5.2.

5.1 Estimation approach

To take the model to the data, I allow users, indexed by i, to differ in their drama values δ_i, temptation durations χ_i, and initial token endowments a_{i1}. For each user, I observe the unlocking history $\boldsymbol{d}_i$, viewing rounds $\boldsymbol{r}_i$, habit stock $\boldsymbol{h}_i$, token balance $\boldsymbol{a}_i$, and, whenever a top-up occurs, the chosen package $\boldsymbol{k}_i^*$ and the corresponding token amount $\boldsymbol{b}_{ik^*}$. I first define the estimator and then discuss the main sources of identification.

Estimator. The parameter vector is

$$\boldsymbol{\theta} := \{ \underbrace{\mu_\delta, \sigma_\delta, \sigma_\eta, \overline{\psi}, \beta_\epsilon}_{\text{intrinsic value and learning}}, \underbrace{\kappa, \alpha_\chi, \pi_0}_{\text{temptation}}, \underbrace{\alpha_1, \alpha_2, \rho}_{\text{habit}}, \underbrace{\omega}_{\text{price sensitivity}} \}.$$

The parameters μ_δ and σ_δ govern the distribution of latent drama value, $\delta_i \sim \mathcal{N}(\mu_\delta, \sigma_\delta^2)$. The learning signal has standard deviation σ_η, so each simulated user receives an individual signal path and updates her posterior belief $\widehat{\delta}_{it}$ over time. The parameter $\overline{\psi}$ captures the nighttime outside option, and β_ϵ governs the dispersion of the episode-level unlocking taste shock. Temptation is characterized by its per-minute magnitude, κ, and its effective duration distribution. The duration distribution has a point mass π_0 at $\chi = 0$, interpreted as zero effective rolling temptation, and, conditional on positive effective temptation, follows a discrete Pareto distribution on $\{1, \ldots, 80\}$ with shape parameter α_χ; all longer durations are censored at $\chi = 80$.[37] Habit formation is governed by the quadratic utility parameters (α_1, α_2) and by ρ, the probability that a viewing round ends after an episode. Finally, ω captures price sensitivity. I normalize the scale of the top-up taste shock by setting $\beta_\nu = 1$, so that the shock follows a standard Gumbel distribution.

Because user-specific values δ_i are unobserved, maximum likelihood would require integrating over latent heterogeneity and simulated belief paths, which is computationally costly in this dynamic setting. I therefore estimate the model using the Simulated Method of Moments (SMM), minimizing

$$\widehat{\boldsymbol{\theta}}_{SMM} = \arg\min_{\boldsymbol{\theta}} \left(\widehat{\boldsymbol{m}} - \widehat{\boldsymbol{m}}^S(\boldsymbol{\theta})\right)' \boldsymbol{W} \left(\widehat{\boldsymbol{m}} - \widehat{\boldsymbol{m}}^S(\boldsymbol{\theta})\right), \tag{17}$$

where $\widehat{\boldsymbol{m}}$ denotes moments constructed from the observed data and $\widehat{\boldsymbol{m}}^S(\boldsymbol{\theta})$ denotes the average of the corresponding moments across S simulations of model-implied user behavior (McFadden, 1989; Pakes and Pollard, 1989). The weighting matrix $\boldsymbol{W}$ is constructed from bootstrap variation in the empirical moments, following Agarwal (2015). In each simulation, I simulate users' top-up and unlocking decisions, conditioning on their initial token endowments a_{i1} and the observed portion of their round histories. I draw user-specific drama values, temptation durations, and episode-level learning signals, and then update posterior beliefs $\widehat{\delta}_{it}$ recursively along each simulated path. To reduce sensitivity to starting values and local optima in (17), I first conduct grid searches and then refine the estimates using a local gradient-free Nelder–Mead algorithm. Appendix D.2 reports the estimation routines and intermediate results.

I target four categories of moments: habit stock, top-up package purchases, unlocking behavior, and learning-related top-up dynamics. For habit stock, I target the probability of stopping after episode 9 among users with zero initial endowment, separately for users whose last habit stock satisfies $h_9 \leq 4$ and $h_9 \geq 5$. I also include average habit stock conditional on purchasing

[37] The quantitative results are robust to alternative parameterizations of the duration distribution.

each top-up package. For top-up choices, I target sales of each package across episode bins 10–20, 21–40, 41–60, and 61–80, capturing both the level and evolution of package choices. For unlocking, I target the share of users unlocking episodes 2 and 3, as well as episode groups 4–9, 10–14, and 15–80. These ranges capture the most informative changes in viewership dynamics, as shown in Figure 3a. I also target the fraction of viewing that occurs at night. Finally, to discipline learning and temptation, I include moments describing repeat and upgrade rates after users' kth \$4.99 purchase, as summarized in Figure 4e. Appendix D.2 provides the complete list of moments. All parameters are estimated jointly; the discussion below describes which moments are most informative about each economic object.

Intuition for identification. I begin with temptation and its separability from learning. In the model, positive effective temptation is the only source of time inconsistency, so the most informative moments are those in which users choose low-commitment packages and then repeatedly continue. The positive market share of small top-up packages, together with repeated purchasing behavior, disciplines the magnitude of per-minute temptation, κ. Package-size choices also inform the distribution of positive effective temptation durations, α_χ: users with longer horizons anticipate more locally tempting continuation and are therefore more likely to choose larger packages. The mass at $\chi = 0$, π_0, is disciplined by the share of users whose choices are consistent with zero effective rolling temptation. The sharpest moments for separating temptation from learning are the repeated top-up transitions documented in Figures 4d and 4e. A learning-only model predicts sorting toward upgrading or stopping as information accumulates, whereas renewed temptation predicts persistent repurchase of small packages.

Habit formation, governed by (α_1, α_2), is identified from variation in habit stock. Users with higher habit stock are empirically more likely to continue watching, making stopping probabilities at different habit-stock levels informative about habit formation.[38] Habit formation also affects continuation values and therefore top-up package choices. Average habit stock conditional on each top-up choice helps discipline (α_1, α_2), while the evolution of habit stock across viewing rounds informs ρ, the probability of losing habit stock after an episode.

The learning parameter σ_η is disciplined by choice dynamics after users have observed early episodes and made an initial small-package purchase. Additional signals should update beliefs and sort users toward either stopping or larger packages as uncertainty resolves. Repeat and upgrade moments after early \$4.99 purchases therefore help distinguish uncertainty about drama quality from persistent self-control problems. The intrinsic-value distribution, $(\mu_\delta, \sigma_\delta)$, is disciplined by residual variation in unlocking and top-up choices after accounting for temptation, learning, and habit formation. As discussed in Section 4.2, users with higher δ are more likely to purchase the \$29.99 package and complete the series, whereas users with lower δ are more likely to stop. Thus, completion rates, outside-option behavior, and the market share of the largest package provide information about $(\mu_\delta, \sigma_\delta)$. The nighttime outside value $\overline{\psi}$ is informed by the share of viewing that occurs at night. The unlocking shock parameter β_ϵ is disciplined by overall

[38] Appendix C.2 provides a fuller discussion of the related empirical patterns.

Table 5: Estimation: SMM estimators and standard error

Parameter	Estimator	Std	Meaning
1. Intrinsic value			
μ_δ	-0.020	0.010	Mean of latent drama value distribution
σ_δ	0.093	0.030	Standard deviation of latent drama value distribution
σ_η	0.039	0.0075	Standard deviation of episode-level learning signal
β_ϵ	0.046	0.0082	Magnitude of unlocking taste shifter
$\overline{\psi}$	0.019	0.057	Outside value during nighttime
2. Temptation			
κ	0.211	0.105	Magnitude of naive temptation
α_χ	0.516	0.107	Pareto shape of positive temptation duration
π_0	0.288	0.090	Mass with zero effective rolling temptation
3. Habit formation			
α_1	$-1.40\text{e-}5$	1.26e-6	Habit formation utility $\alpha(h) = \alpha_1 h (h - \alpha_2)$
α_2	1103.87	3.48e-5	Habit formation utility $\alpha(h) = \alpha_1 h (h - \alpha_2)$
ρ	0.046	0.0093	Probability of entering next round after watching
4. Price sensitivity			
ω	0.902	0.105	Price sensitivity

Notes: The GMM robust standard errors are reported in the table. The habit-formation parameters are structural parameters of $\alpha(h) = \alpha_1 h(h - \alpha_2)$; Figure 7d reports the economically interpretable implied habit function over the observed support.

viewership dynamics, since larger shocks generate more idiosyncratic stopping behavior.

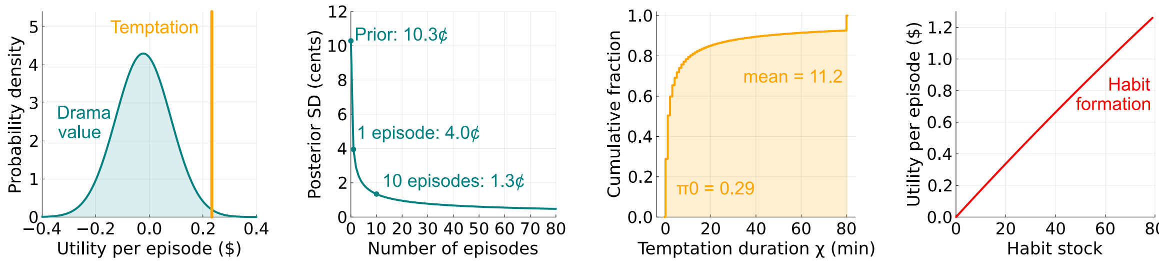
Finally, price sensitivity ω is identified from top-up choices involving real monetary payments. The four packages differ in observed prices and token amounts, and promotional variation shifts token quantities within price points. These sources of variation make package market shares informative about users' price sensitivity. Because top-up options differ only in observed prices and token quantities, the setting also limits the usual concern that unobserved product quality is correlated with price across alternatives in the same choice set.[39]

5.2 Results

I report the estimation results for $\widehat{\boldsymbol{\theta}}_{SMM}$ in Table 5 and graphically in Figure 7. The targeted moments are displayed in Appendix D.3. The estimated price sensitivity is $\omega = 0.90$, which translates utility into real monetary terms (USD). I use this numéraire to report all quantitative results.

The first set of parameters describes the distribution of intrinsic value, which is also represented in Figure 7a. The estimated per-episode drama value δ has a mean of -0.020 (-2.2¢) and a standard deviation of 0.093 (10.3¢). The outside value during nighttime is 0.019 (2.1¢) per minute, whereas the daytime outside value is normalized to 0. The idiosyncratic taste shifter ϵ has a scale

[39] In traditional markets, such as used cars, identifying price sensitivity is complicated by omitted product attributes: buyers may observe quality dimensions that econometricians do not, and these attributes affect both prices and choices. In this setting, top-up options differ only in two observed dimensions—price and number of tokens—which sharply limits unobserved quality differences across alternatives conditional on the choice set.



(a) Value (b) Learning (c) Temptation duration (d) Habit formation

Figure 7: Estimation: Intrinsic value, learning, temptation, and habit formation

Notes: Panel (a) shows the estimated distribution of latent drama value δ (teal curve) alongside the temptation level (orange bar), all normalized in dollars using the estimated price sensitivity. Panel (b) shows the decline in posterior uncertainty about δ as users observe episodes. Panel (c) displays the cumulative distribution of effective temptation duration χ, which has a mass at $\chi = 0$ and a positive Pareto tail capped at $\chi = 80$. Panel (d) illustrates the estimated function for habit formation $\alpha(h)$, which is nearly linear with respect to users' habit stock.

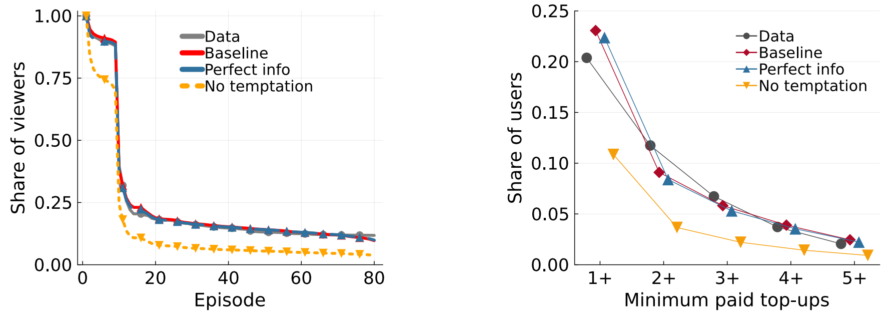
parameter of 0.046, resulting in a standard deviation of 0.059 (6.6¢). These results indicate that, on average, users experience a net loss of 2.8¢ per one-minute episode compared with their outside options, highlighting the potential loss in user surplus.[40]

The learning signal has a standard deviation of 0.039 (4.3¢), implying that users receive informative feedback about the latent value of a drama from each episode. Figure 7b suggests that this learning process is rapid: after the first episode, uncertainty about drama quality falls by 61%, and declines further to 1.3¢ by the time users make their first top-up choice. Learning therefore remains part of the structural environment, but slow resolution of quality uncertainty is unlikely to be the main force behind repeated paid top-ups.

Temptation is estimated at 0.211 (23.3¢) per minute, placing it at the 99th percentile of the drama value distribution. As shown in Figure 7c, 28.8% of users have zero effective rolling temptation, while the remaining users are drawn from a heavy-tailed positive Pareto distribution capped at 80 minutes. The average effective temptation horizon is 11.2 minutes, with a standard deviation of 22.6 minutes. Approximately 22.9% of users have effective temptation horizons longer than 8 minutes (the duration covered by the \$4.99 package), 16.0% exceed 17 minutes (covered by the \$9.99 package), and 10.9% exceed 37 minutes (covered by the \$19.99 package). The cap at $\chi = 80$ absorbs 7.4% of users. These estimates imply a short-run force that is short in clock time but long relative to one-minute episodes for users with positive effective temptation.

I also estimate substantial habit formation, as shown in Figure 7d. Rather than interpreting the extreme raw support parameter α_1 and α_2 directly, I focus on the implied habit function over the observed support. The estimated function is nearly linear in habit stock and ranges from 0 to \$1.27 per episode. Watching an additional episode in the round contributes, on average, 1.6¢ in per-episode utility through habit formation. This habit channel helps explain high willingness

[40]The average loss from watching one episode equals its average value of -2.2¢ minus the average value of the outside option, 0.27×2.1¢.



(a) Viewership (b) Repeat top-ups

Figure 8: Temptation predicts persistence in viewership and top-ups

Notes: The figure compares observed data, the estimated baseline model, a perfect-information counterfactual that sets $\sigma_\eta = 0.0001$, and a no-temptation counterfactual that sets temptation duration to zero, holding the remaining estimated environment fixed. Panel (a) reports the share of viewers remaining by episode. Panel (b) reports the share of first-episode viewers with at least a given number of paid top-ups.

to pay among users who continue, but it has a different welfare interpretation from temptation because it enters the long-run welfare utility. The estimated probability of a round change, ρ, is 4.6%, influencing users' decisions through their expectations of future habit formation.

5.3 Model decomposition: temptation and learning in persistent behavior

Figure 8 provides a decomposition exercise for the temptation mechanism by comparing the estimated model with two counterfactual environments. The first removes temptation while holding the rest of the estimated environment fixed. The second makes information perfect, while preserving the estimated temptation and habit-formation components. This comparison is useful because continuation and repeat top-up behavior could, in principle, be generated either by a short-run temptation force that induces persistence even when the rational surplus is low or by users gradually learning that a drama is valuable.

Panel (a) shows that removing temptation sharply reduces retention throughout the episode sequence, while the perfect-information economy remains close to the baseline. Panel (b) gives the same message on the payment margin. In the data, 20.4% of viewers make at least one paid top-up and 11.7% make at least two. The baseline model predicts 23.1% and 9.1%, respectively, whereas the no-temptation counterfactual predicts only 10.9% and 3.7%. By contrast, the perfect-information counterfactual remains close to the baseline, predicting 22.4% and 8.4%. This comparison suggests that slow learning about drama quality alone is unlikely to explain persistence in paid top-ups. Under the structural model, the estimated temptation component is needed to match continued viewing and repeated paid purchases.

6 Main results

In this section, I present the model's quantitative implications for product granularity, temptation, and user surplus. I begin in Section 6.1 by defining the temptation-induced surplus loss. Section 6.2 studies the comparative static over episode length. Section 6.3 evaluates counterfactual design changes that alter the size or timing of stopping decisions.

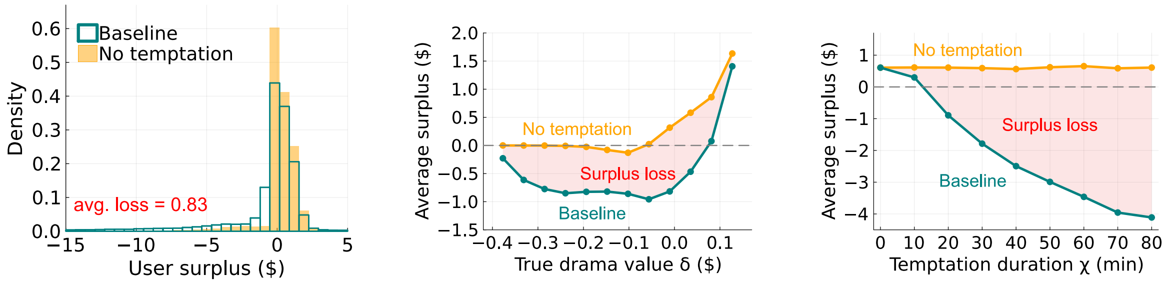
6.1 Surplus loss under the long-run welfare criterion

I evaluate welfare under the model's long-run welfare criterion. User surplus includes intrinsic content value and habit formation according to equation (8), net of token payments; the temptation term affects choice utility but is excluded from welfare.[41] This is a maintained normative criterion, not a directly observed preference object. I also net out the utility from the first episode, which is mechanically consumed in the model, so the welfare measure focuses on endogenous continuation and top-up decisions. I therefore interpret the welfare exercises as temptation-induced surplus changes under this criterion.

Under this criterion, the no-temptation counterfactual isolates continuation generated by the estimated temptation component. Figure 8a illustrates the share of remaining viewers per episode. The baseline model prediction (red line) closely tracks the observed data pattern (gray line). Removing temptation results in a substantial decline in viewer retention across episodes. For instance, without temptation, 16.4% of users stop watching after the first episode, versus 5.4% in the baseline. By the final episode, only 3.7% of users would complete the series, a 61.4% reduction from the baseline completion rate of 9.7%. This gap is the behavioral counterpart of the temptation-induced surplus loss under the maintained long-run welfare criterion.

I compute surplus loss using the same counterfactual. As shown in Figure 9a, removing temptation shifts the surplus distribution to the right, reducing the mass of users with large negative surplus. Under the long-run welfare criterion, the average user loses \$0.83 from the estimated temptation component. Figures 9b and 9c show where these losses arise. By drama value, the no-temptation benchmark is close to zero for users with very low δ, because they stop after the free first episode once its utility is netted out. The loss is largest for users near the margin of valuing the drama, for whom the estimated temptation component generates substantial extra consumption that is not justified by welfare-relevant utility, consistent with the inverted-U relationship discussed in Section 4.3. By temptation horizon, the pattern is monotone: users with $\chi = 0$ have no surplus loss by construction, while the gap between the no-temptation benchmark and the baseline widens sharply for users with longer temptation horizons. This pattern confirms the central mechanism of the model under the maintained welfare criterion: temptation sustains engagement precisely for users whose welfare-relevant continuation value is limited.

[41] My analysis is qualitatively robust with other measures of welfare, such as the pure surplus using only the intrinsic value.



(a) Surplus distribution (b) By drama value (c) By duration

Figure 9: User surplus under the long-run welfare criterion

Notes: I compare the baseline model with a no-temptation counterfactual obtained by setting $\kappa = 0$, holding the estimated learning, habit-formation, and top-up environments fixed. User surplus excludes temptation and nets out the utility from the mechanically consumed first episode. Panel (a) plots the distribution of user surplus in the baseline model (teal outline) and in the no-temptation counterfactual (light orange). Panels (b) and (c) plot average surplus in the baseline model (teal) and in the no-temptation counterfactual (orange), respectively by true drama value δ and temptation horizon χ. The light-red shaded area is the temptation-induced surplus loss under the maintained long-run welfare criterion.

6.2 Short length amplifies self-control problems

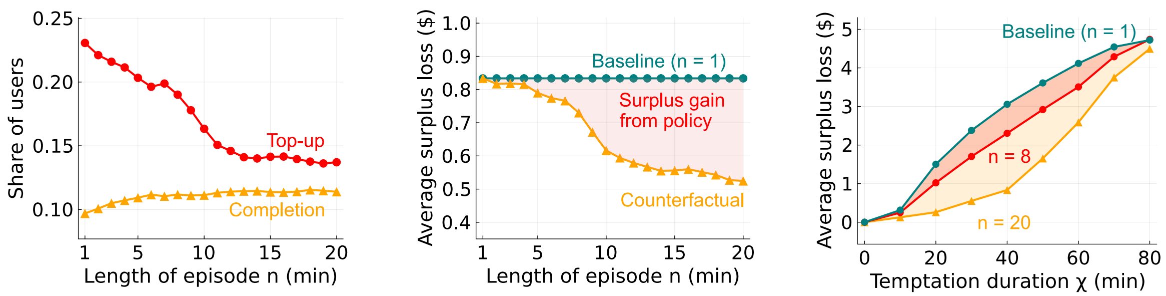
I now explore the mechanism by which short episode length amplifies self-control problems, using a comparative static exercise that varies episode duration. To avoid confounding effects from the platform's dynamic pricing rule, I hold the length of the first nine free episodes fixed. For subsequent episodes, let $n \in \mathbb{N}$ denote the counterfactual episode length, effectively merging n consecutive one-minute episodes into a single, longer episode. Let τ index these counterfactual episodes, with each τ corresponding to a set $Q_\tau \subset \{1, \ldots, T\}$ of original episodes. As in the stylized model in Section 2, I assume that once an episode is unlocked, the user watches it in full. The perceived flow utility is given by:

$$\widetilde{u}_\tau(h, \delta, \psi, \chi, \epsilon; n) = \mathbb{E}_\tau \left[n\delta - \sum_{t \in Q_\tau} \psi_t + \sum_{t \in Q_\tau} \alpha(h_t) + \min\{\chi, n\}\kappa + \sum_{t \in Q_\tau} \epsilon_t \right], \tag{18}$$

where the expectation is taken with respect to the information available in the first original episode of Q_τ, including habit stock, outside option, and taste shifter.[42] As in the stylized model, the key mechanism is a reduction in temptation due to the $\min\{\chi, n\}$ term, which dampens the effect of short-term bias. The counterfactual model is solved analogously to the baseline, using the same top-up framework.

Figure 10 reports the behavioral and welfare consequences of increasing episode length. Panel (a) first shows the behavioral margin. As n rises from 1 to 20 minutes, the share of users who make any top-up falls from 23.1% to 13.7%, while the completion rate rises modestly from 9.7%

[42] I assume users observe the habit stock, outside value, and taste shifter associated with the first original episode in Q_τ and form beliefs about the rest, ensuring comparability between the counterfactual and baseline scenarios.



(a) User behavior (b) Average surplus loss (c) Loss by temptation duration

Figure 10: Comparative static: Episode length and self-control problems

Notes: These figures illustrate the counterfactual effects of setting a minimum episode length on user behavior and surplus. Panel (a) plots the share of users who top up and the share who complete the series. Panel (b) plots temptation-induced surplus loss, defined as no-temptation surplus minus baseline surplus at the same episode length. The teal line fixes the one-minute benchmark, the orange line shows the counterfactual loss, and the shaded area is the surplus gain from increasing episode length. Panel (c) plots the same surplus-loss object by temptation duration χ: the teal line is the $n = 1$ baseline, while the red and orange lines are the $n = 8$ and $n = 20$ counterfactuals. The red and orange shaded areas show the respective recovered losses and overlap where the gains are common.

to 11.4%. Longer episodes therefore do not simply shut down consumption. Instead, they reduce the payment margin most associated with repeated temptation-driven continuation, while leaving completion approximately stable among users whose continuation value is sufficiently high.

Panel (b) translates this behavioral response into the welfare object used throughout the section: temptation-induced surplus loss, defined as no-temptation surplus minus full-model surplus at the same episode length. The one-minute benchmark implies an average loss of $0.83 per user, consistent with Section 6.1. Increasing episode length to eight minutes lowers this loss to $0.73, recovering $0.10 per user, or 12.4% of the benchmark loss. Extending the visual range to $n = 20$ makes the policy gradient more visible: the counterfactual loss falls to $0.52, recovering $0.31 relative to the one-minute benchmark.[43]

Panel (c) shows that the surplus gain from increasing episode length is hump-shaped in temptation duration. Very low-χ users have little surplus loss to recover, so the gain is small near $\chi = 0$. The gain then rises for lower- and intermediate-duration users: for these consumers, temptation is strong enough to distort continuation decisions in the one-minute baseline, but short enough that bundling episodes meaningfully reduces the frequency with which temptation affects choices. For very high-χ users, the gain declines because temptation remains active even within longer episodes, so increasing n only partially attenuates the self-control problem. This inverted-U pattern mirrors the mechanism in Proposition 1: short episodes are especially harmful when temptation is persistent relative to one-minute clips but not so persistent that longer episodes leave the temptation channel largely intact.

[43]For $n > 8$, the exercise also interacts with the top-up package menu, because the $4.99 package covers about seven one-minute episodes. I therefore use the $n = 8$ change as the main welfare comparison and interpret larger values of n as visualizing the policy gradient.

6.3 Counterfactual design changes

I now study counterfactual design changes that mitigate self-control problems by changing the effective size or timing of stopping decisions. I begin with a platform-specific margin: restrictions on the minimum top-up package size. Because episodes are sold through token bundles, removing small packages increases the amount of content purchased at each top-up and directly targets the low-commitment repurchase margin. I interpret this exercise as a platform-specific mechanism counterfactual for product granularity. I then consider two broader behavioral interventions: a default time limit, motivated by Proposition 2, and a mandatory break following each top-up purchase. Both preserve access while weakening the renewal of local temptation.

Package-size mechanism counterfactual. I first consider a platform-design counterfactual that restricts the minimum size of token packages. Because episodes are bundled and sold through packages, removing the smallest packages increases the effective content commitment at each purchase. This counterfactual isolates the product-granularity mechanism: when small, low-commitment packages are unavailable, users who would otherwise repeatedly make small top-ups must either commit to a larger package or stop. I implement the counterfactual by sequentially removing the \$4.99 package and then both the \$4.99 and \$9.99 packages.

Results are reported in columns (2) and (3) of Table 6. Removing the \$4.99 package raises average user surplus from $-$\$0.25 to \$0.21 and reduces average platform profit from \$5.32 to \$4.01. The \$0.46 surplus gain corresponds to a 55.4% reduction in the baseline temptation-induced surplus loss of \$0.83 per user. The share of users who purchase tokens falls from 23.1% to 14.2%, while average top-up spending among token buyers rises from \$23.05 to \$28.35. Thus, the policy reduces entry into paid consumption but selects remaining token buyers on higher posterior continuation value. Removing both the \$4.99 and \$9.99 packages raises user surplus further to \$0.32, implying a \$0.57 gain and a 68.7% reduction in the baseline temptation-induced loss, while the token-buyer share falls to 12.2%. These results show that the low-commitment repurchase margin is quantitatively important for temptation-driven welfare losses.

Default time limit. I next consider a default time-limit policy tied to the duration implied by the user's first top-up package. By default, users may watch only the free episodes and the number of additional episodes covered by their first top-up. Before viewing begins, they may opt out of the limit by paying a cost valued at Φ dollars.[44] Users who opt out face the baseline problem in Section 4. I therefore solve the restricted problem for users who comply and then compare its value with the value of opting out.

In the model, the time limit is equivalent to restricting compliant users to a single top-up opportunity. I introduce an additional state variable $\ell_t \in \{0, 1\}$ denoting the number of remain-

[44] The opt-out cost can be interpreted as a friction, such as a time delay, valued at Φ, or as a monetary fee charged by the platform to override the limit. Because the model includes random taste shifters, adherence to the time limit requires a positive opt-out cost Φ.

Table 6: Counterfactual design changes: Package size, default time limit, and mandatory break

Outcomes	Baseline	I. Min. Pack. Size		II. Time Limit		III. Break	
		$> \$4.99$	$> \$9.99$	$\Phi = 0.1$	$\Phi = 0.5$	1 min	5 min
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
I. *Per-user surplus*							
User surplus ($)	-0.25	0.21	0.32	-0.16	0.06	-0.20	-0.07
Profit ($)	5.32	4.01	3.64	4.54	3.87	5.09	4.61
II. *User activity*							
Top-up per token buyer ($)	23.05	28.35	29.73	22.47	21.09	23.13	23.14
Frac. token buyer	0.23	0.14	0.12	0.20	0.18	0.22	0.20
Frac. completion	0.10	0.10	0.10	0.10	0.10	0.09	0.09
Frac. time limit complier				0.77	0.99		

Notes: This table presents outcomes under three classes of counterfactual design changes. Column (1) reports the baseline results. Columns (2) and (3) assess package-size restrictions by sequentially removing lower-priced top-up options; I interpret these as mechanism counterfactuals for product granularity rather than preferred policy recommendations. Columns (4) and (5) evaluate default time limits with increasing opt-out costs Φ in dollars. Columns (6) and (7) analyze mandatory breaks at top-up with different break lengths. For each scenario, I report average user surplus under the long-run welfare criterion, platform profit, average top-up spending per token purchaser, the share of users who purchase tokens, the share who complete the series, and, where applicable, the share who comply with the time limit.

ing top-up opportunities. The state is therefore $(h_t, a_t, \widehat{\delta}_t, \ell_t)$, where $\widehat{\delta}_t$ evolves according to the Bayesian updating rule in equation (10). The value functions under the time-limit policy, denoted by superscript TL, are

$$\begin{aligned} V_t^{TL}(\mathbf{s}, \ell; \psi, \chi, \boldsymbol{b}, \boldsymbol{\nu}) = \quad & \mathbb{1}\{a < c_t\}\mathbb{1}\{\ell > 0\} \max_{k \in \{0,\ldots,K\}} \left\{ EW_t^{TL}(h, a + b_k, \widehat{\delta}, \ell - 1; \psi, \chi) - \omega p_k + \nu_k \right\} \\ & + \mathbb{1}\{a \geq c_t\} EW_t^{TL}(h, a, \widehat{\delta}, \ell; \psi, \chi), \qquad (19) \\ EV_t^{TL}(\mathbf{s}, \ell; \psi, \chi) = \quad & \mathbb{1}\{a < c_t\}\mathbb{1}\{\ell > 0\} \mathbb{E}_{\boldsymbol{b}} \left[\log \left(\sum_{k=0}^{K} e^{EW_t^{TL}(h, a + b_k, \widehat{\delta}, \ell - 1; \psi, \chi) - \omega p_k} \right) \right] \\ & + \mathbb{1}\{a \geq c_t\} EW_t^{TL}(h, a, \widehat{\delta}, \ell; \psi, \chi). \qquad (20) \end{aligned}$$

When $a < c_t$ and $\ell = 0$, the user can no longer top up, and the value of the outside option is normalized to zero. The functions $W_t^{TL}(\cdot)$ and $EW_t^{TL}(\cdot)$ are defined analogously to equations (15) and (16), replacing $EV_t(\cdot)$ with $EV_t^{TL}(\cdot)$ and taking expectations over $\widehat{\delta}'$ conditional on the current posterior mean $\widehat{\delta}$. The key restriction is $\ell_1 = 1$: compliant users have only one top-up opportunity.

A user with initial endowment a_1, prior mean $\widehat{\delta}_1 = \mu_\delta$, temptation duration χ, and initial outside option ψ_1 opts out if and only if

$$\underbrace{EV_1^{TL}(0, a_1, \widehat{\delta}_1, 1; \psi_1, \chi)}_{\text{Expected value with time limit}} < \underbrace{EV_1(0, a_1, \widehat{\delta}_1; \psi_1, \chi)}_{\text{Expected value without limit}} - \underbrace{\omega\Phi}_{\text{Opt-out cost in utility units}} . \qquad (21)$$

The policy is most valuable for users who expect limited continuation value but would repeatedly purchase small packages when temptation renews. Users with high continuation value are more likely to opt out or purchase large packages and complete the series regardless of the restriction.

Columns (4) and (5) of Table 6 report the results. A small opt-out cost of \$0.10 raises average user surplus from −\$0.25 to −\$0.16 under the long-run welfare criterion. This \$0.09 gain reduces the baseline temptation-induced surplus loss by 10.8%. Platform profit falls from \$5.32 to \$4.54 because the policy reduces top-up purchases on both the intensive and extensive margins. Average top-up spending among token buyers falls from \$23.05 to \$22.47, and the token-buyer share falls from 23.1% to 20.0%. When the opt-out cost rises to \$0.50, the complier share increases from 77.2% to 98.7%, average user surplus rises to \$0.06, and platform profit falls to \$3.87. The implied \$0.31 surplus gain corresponds to a 37.3% reduction in the baseline temptation-induced loss.

Mandatory break during top-ups. Finally, I consider a mandatory break following each top-up purchase. The implementation requires users to wait through an ad of ϕ minutes before resuming the drama. During the break, users receive their outside value, derive no intrinsic utility from the ad, and do not feel tempted to continue watching it.[45] The break therefore shortens the effective temptation horizon at the moment of the top-up decision.

I incorporate the break by modifying the top-up value functions in equations (12) and (13). When the user considers purchasing a package, the continuation value is evaluated at the reduced temptation duration $\max\{\chi - \phi, 0\}$:

$$
\begin{aligned}
V_t^B(\mathbf{s};\psi,\chi,\boldsymbol{b},\boldsymbol{\nu}) = \quad & \mathbb{1}\{a < c_t\} \max_{k\in\{0,\ldots,K\}} \left\{ EW_t(h, a+b_k, \widehat{\delta}; \psi, \max\{\chi-\phi,0\}) - \omega p_k + \nu_k \right\} \\
& + \mathbb{1}\{a \geq c_t\} EW_t(h, a, \widehat{\delta}; \psi, \chi), \qquad (22) \\
EV_t^B(\mathbf{s};\psi,\chi) = \quad & \mathbb{1}\{a < c_t\} \mathbb{E}_{\boldsymbol{b}} \left[\log\left(\sum_{k=0}^{K} e^{EW_t(h,a+b_k,\widehat{\delta};\psi,\max\{\chi-\phi,0\})-\omega p_k} \right) \right] \\
& + \mathbb{1}\{a \geq c_t\} EW_t(h, a, \widehat{\delta}; \psi, \chi). \qquad (23)
\end{aligned}
$$

Together with the unlocking value functions (15) and (16), the belief transition (10), and the terminal condition (11), these equations define the counterfactual choice problem. The only change from the baseline is that the perceived temptation horizon is reduced at top-up decisions.

Columns (6) and (7) of Table 6 report the results. A one-minute break raises average user surplus from −\$0.25 to −\$0.20 and reduces the token-buyer share from 23.1% to 22.0%. The \$0.05 surplus gain corresponds to a 6.0% reduction in the baseline temptation-induced surplus loss. A five-minute break raises user surplus further to −\$0.07 and lowers the token-buyer share to 19.9%. This \$0.18 gain reduces the baseline temptation-induced loss by 21.7%. The welfare gains are smaller than under the package-size and default-limit counterfactuals, but they are positive and arise from the same mechanism: weakening the renewal of temptation at the moment users decide whether to continue paying.

[45] The assumption that users receive their outside value during ads reflects the possibility that they engage in other activities while ads play, so the break does not mechanically lower welfare. If users actively dislike ads, the counterfactual would impose an additional cost not modeled here.

7 Extension: From Short Dramas to Short Videos

This section extrapolates the estimated temptation structure to a stylized short-video environment. The exercise is a calibration, not an estimate of welfare on TikTok or any specific platform. It asks how large the product-granularity mechanism could be if the temptation magnitude and temptation-horizon distribution estimated from paid short dramas transferred to unpaid short-video consumption. The exercise abstracts from payments, habit formation, learning, social interaction, algorithmic targeting, platform entry and exit, and differences in content supply. I therefore interpret the results as scale benchmarks under explicit transfer assumptions.

The full-transfer benchmark returns to the stylized model in Section 2. I use the estimated temptation magnitude κ, the estimated distribution of net intrinsic value $x = \delta - \psi$, and the estimated distribution of temptation horizons χ from Table 5 and Figure 7. Users face a one-hour short-video environment with one-minute content units. The welfare criterion is the same long-run criterion used in the structural model.

Full-transfer benchmark. In the one-hour benchmark, average surplus is $-\$0.44$ per user. Approximately 19.2% of simulated users choose not to watch. Eliminating temptation raises average hourly surplus to \$1.72, implying a temptation-induced surplus loss of \$2.16 per hour. The loss is concentrated: 41.6% of users experience a positive welfare loss, and the average loss among affected users is \$5.19 per hour. The distribution of user surplus is reported in Figure 11a. To translate these hourly quantities into a scale benchmark, I use 170 million U.S. TikTok users and 29.2 monthly hours of usage.[46] Because the simulated hourly loss is averaged over all model users, including the 19.2% who do not watch, I convert it to an active-user scale using the model active share of 80.8%. The implied monthly loss is

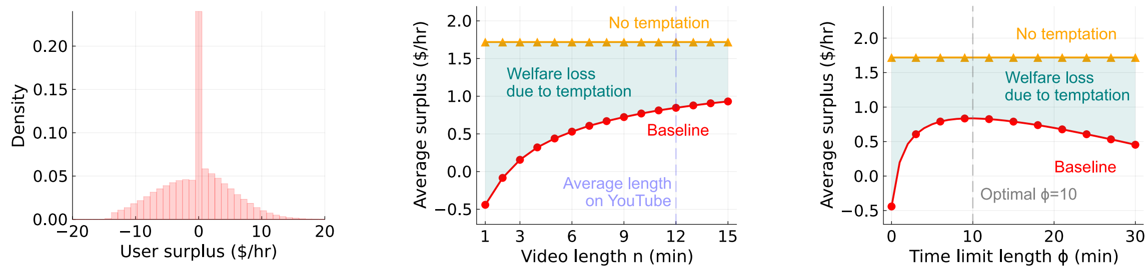
$$170 \text{ million} \times 29.2 \times \frac{\$2.16}{0.808} = \$13.3 \text{ billion}.$$

This number is not an estimate of TikTok welfare. It is the full-transfer scale benchmark for the estimated product-granularity mechanism.

Figure 11b implies that short content length is quantitatively important. Increasing the content unit from one minute to 12 minutes, which is the average length of videos on YouTube, raises hourly surplus from $-\$0.44$ to \$0.85. Relative to the no-temptation benchmark of \$1.72, this reduces the temptation-induced surplus loss from \$2.16 to \$0.87 per hour. The longer content unit therefore recovers \$1.29 per hour, or 59.6% of the one-minute temptation loss. At the aggregate scale above, this corresponds to \$7.9 billion per month.

I also evaluate a default time-limit counterfactual. As in Proposition 2, users with temptation horizon $\chi \leq \phi$ comply with a ϕ-minute default and stop at the limit, while users with $\chi > \phi$

[46] The user-count input is 170 million U.S. TikTok users, based on the Supreme Court's January 17, 2025 opinion in *TikTok Inc. v. Garland*. The time-use input is 58.4 minutes per day, or 29.2 hours per month, from the eMarketer source cited in the draft.



(a) Distribution of hourly surplus (b) Increasing video length n (c) Default time limit ϕ

Figure 11: Calibrated hourly surplus in a short-video environment

Notes: Panel (a) shows the distribution of calibrated hourly surplus from short-video consumption. The spike at zero reflects the 19.2% of simulated users who choose not to watch. Panels (b) and (c) report the effects of increasing content length n and introducing a default time limit ϕ, respectively. The red curve shows average surplus with temptation, the orange curve shows the no-temptation benchmark, and the shaded region represents temptation-induced surplus loss under the transferred long-run welfare criterion.

opt out and consume the full hour. Average surplus is inverted-U shaped in the default length: a longer limit covers more users but allows each complier to consume more before stopping. Figure 11c reports the calibration results. The optimal default is 10 minutes, which raises hourly surplus from $-\$0.44$ to $\$0.84$. Equivalently, it reduces the temptation-induced surplus loss from \$2.16 to \$0.88 per hour, recovering \$1.28 per hour, or 59.1% of the one-minute loss. At scale, this corresponds to \$7.8 billion per month.

The preceding calculations report consumer-surplus gains. In unpaid short-video markets, however, reduced viewing may also reduce platform and advertiser surplus. Rather than model the advertising side directly, I compute break-even revenue thresholds. Let ΔCS_p denote the consumer-surplus gain from counterfactual p, and let ΔT_p denote the number of viewing hours displaced per baseline viewing hour. If r is platform plus advertiser surplus per viewing hour, the total-surplus effect is $\Delta W_p = \Delta CS_p - r\Delta T_p$. The counterfactual remains total-surplus-improving whenever $r < r_p^* \equiv \Delta CS_p / \Delta T_p$. For the 12-minute content-unit counterfactual, the consumer-surplus gain is \$1.29 per hour and displaced viewing equals 0.196 hours per baseline hour. The break-even value is therefore \$6.56 per displaced viewing hour. For the 10-minute default, the consumer-surplus gain is \$1.28 per hour and displaced viewing equals 0.246 hours per baseline hour, implying a break-even value of \$5.19 per displaced viewing hour. Thus, the counterfactuals remain total-surplus-improving unless platform plus advertiser surplus from displaced viewing exceeds these thresholds. This calculation makes transparent how large omitted platform-side surplus would need to be to overturn the consumer-surplus gains.

Sensitivity and interpretation. The full-transfer benchmark is precise but assumption-heavy. I therefore report a conservative stress test that weakens the transfer assumptions along three margins. First, only half of TikTok users are assumed to be behaviorally comparable to short-drama users. Second, the temptation magnitude is scaled down by 50%. Third, temptation horizons are also scaled down by 50%. Under this stress test, the temptation-induced loss among comparable users is \$0.68 per hour. Aggregating only over the comparable half of users gives a monthly loss of \$2.8 billion. A 12-minute content unit recovers \$0.43 per hour among comparable users, or \$1.8 billion per month. The optimal default remains 10 minutes and recovers \$0.40 per hour, or \$1.7 billion per month. These values should not be interpreted as formal lower bounds. They show that the welfare stakes remain economically large even when the population, magnitude, and duration transfer assumptions are all weakened substantially.

The calibration ladder supports two conclusions. First, the interaction between temptation duration and content length can generate large welfare stakes in short-video environments. The key object is not only whether users are tempted, but whether the temptation horizon is long relative to the content unit. Second, design changes that alter the effective size or timing of stopping decisions can recover a substantial share of the calibrated welfare loss. The appendix reports additional sensitivity exercises that vary population comparability, temptation transfer, external scale inputs, default compliance, session structure, and platform revenue.

8 Conclusion

This paper's main message is that product granularity can transform short-run temptation into persistent demand. When the unit of consumption is shorter than the duration of temptation, each stopping decision arrives while the next unit remains locally tempting. The same preference distortion can therefore have very different welfare consequences depending on how the product is divided and sold. This insight changes how we should interpret high engagement on digital platforms: usage is not sufficient to distinguish high consumer value from distorted continuation. For managers, the implication is that content length, package menus, defaults, and breaks are not merely interface choices. They are demand-design variables that shape the frequency and size of stopping decisions, and therefore determine whether engagement reflects value creation or repeated failures to stop.

Several directions remain open. The extrapolation to short-video platforms illustrates the potential scale of the mechanism, but it also highlights the need for direct evidence in unpaid, algorithmic-feed environments. A natural next step is to run field experiments that randomize content length, default limits, break timing, or bundle size, while measuring both realized consumption and ex-ante plans. Such experiments would help separate the causal effect of product granularity from content selection, recommendation algorithms, social interaction, and platform-side advertising incentives. More broadly, combining revealed-preference data with experimental variation would allow future work to estimate temptation duration directly in other digital markets and evaluate whether welfare-improving design can be aligned with platform profitability.

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Appendix

Appendix A Model

Appendix A.1 Represent temptation with present-biased preferences

In this section, I demonstrate my model of temptation can be micro-founded by present bias within a class of standard quasi-hyperbolic discounting models. This formulation shows that the economic insights derived from my baseline model can be generalized within the classic present-biased preference framework.

Consider a large set of one-minute videos $\{1, ..., T\}$, where each minute represents a unit of time. At any given moment t, the agent chooses between watching videos and working (the outside option). Per-minute work provides delayed utility b after period $t + \chi + 1$, whereas watching a video yields immediate utility $b + x$. Consistent with the baseline model, x captures the intrinsic utility of watching a video relative to the agent's outside option.

Regarding time preferences, I assume the agent is present-biased, with the present defined as a window of χ minutes:

$$U_t = \sum_{\tau=t}^{t+\chi} u_\tau + \beta \sum_{\tau=t+\chi+1}^{T} u_\tau, \tag{A.1}$$

where the present-bias parameter $\beta < 1$ applies to all utility realized after χ minutes. This utility specification represents a quasi-hyperbolic discounting framework where the regular discount factor is set to 1 for clarity.

Given the preference structure in (A.1), the set of users experiencing self-control problems can be characterized as:

$$-(1-\beta)b \leq x \leq 0. \tag{A.2}$$

Agents with $x \leq 0$ would not watch videos under the long-run criterion. However, due to present bias, those with $x \geq -(1-\beta)b$ find video-watching more appealing because $b + x \geq \beta b$, making them willing to watch videos over the current χ-minute horizon. These users experience a self-control problem because their bias continually shifts toward the next set of future videos, leading realized consumption to exceed the current plan.

This quasi-hyperbolic discounting specification serves as the micro foundation for my baseline model of temptation. The temptation duration χ can be equivalently interpreted as the span of present bias, whereas the magnitude of temptation κ is micro-founded in present bias, taking the value $(1-\beta)b$. This equivalence demonstrates that the economic insights derived from my baseline model are directly applicable to the standard present-biased preference framework.

Appendix A.2 Solution for sophisticated users

Agents might not be fully naïve about their temptation. In standard present-bias language, sophistication means that users anticipate their future self-control problem; it does not mean that temptation disappears. Let $\widetilde{\kappa} \in [0, \kappa]$ be the anticipated component of temptation, where $\widetilde{\kappa} = \kappa$ corresponds to full anticipation and $\widetilde{\kappa} \in (0, \kappa)$ corresponds to partial anticipation. The perceived flow utility (1) can be generalized into:

$$\widetilde{u}_{\tau}^{t}(x, \chi) = \underbrace{x}_{\text{intrinsic utility}} + \underbrace{\mathbb{1}\{\tau < t + \chi\}\, \widetilde{\kappa}}_{\text{perceived temptation}} + \underbrace{\mathbb{1}\{\tau < t + \chi\}\, (\kappa - \widetilde{\kappa})}_{\text{unperceived temptation}}. \tag{A.3}$$

For users who are susceptible to self-control problems ($x \in (-\kappa, 0)$), I consider the following two situations:

1. **Users with $x \in (-\widetilde{\kappa}, 0)$ decide not to watch.** Because temptation $\widetilde{\kappa}$ is perceived, these users anticipate that once they begin, the irrational temptation will persist for the next χ videos, preventing them from stopping. They thus evaluate their total utility as $\chi(x + \kappa) + (T - \chi)x < 0$ and choose not to watch.

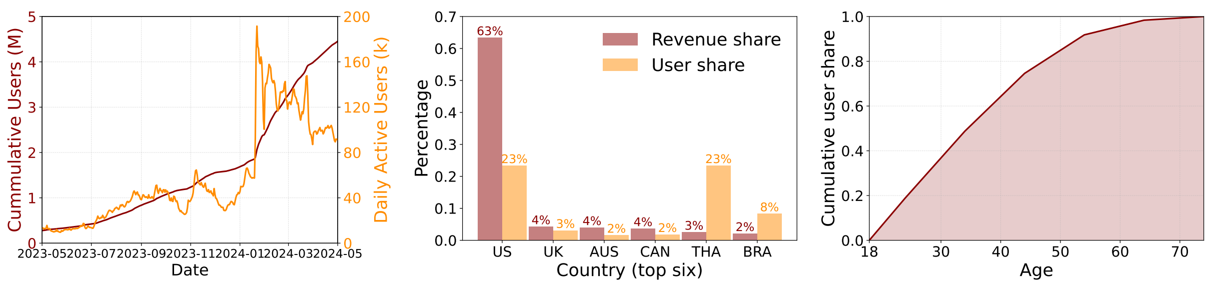
2. **Users with $x \in (-\kappa, -\widetilde{\kappa})$ fall into self-control problems.** Although these users recognize the temptation $\widetilde{\kappa}$, they do not expect to keep watching beyond the temptation duration χ, since the perceived temptation is insufficient to justify continuation. Yet the unperceived component is strong enough to trigger impulsive consumption, resulting in the same self-control problem described above.

The general takeaway from this stylized environment is that anticipation of temptation can diminish self-control problems. In the simple example, full anticipation eliminates overconsumption for users with negative long-run value.

Appendix B Background

I provide further details on the platform in this section, including the descriptive facts on users, dramas, revenue, pricing, and the supply-side details.

Users. The platform size and user characteristics are summarized in Figure B.1. According to panel (a), the total number of users has reached 4 million worldwide and the Daily Active Users (DAU) remains around 100,000 by the May of 2024. Figure B.1b shows a large fraction of users are from US (23%), Thailand (23%), and Brazil (8%), whereas most of the platform revenue is contributed by US users (63%). Country boundaries create a natural form of market segmentation, and I therefore focus on the biggest market, the US market, in our main analysis. Panel B.1c indicates users of this platform are mainly young people, with 36% aged below 30, 63% below 40, and 85% below 50.

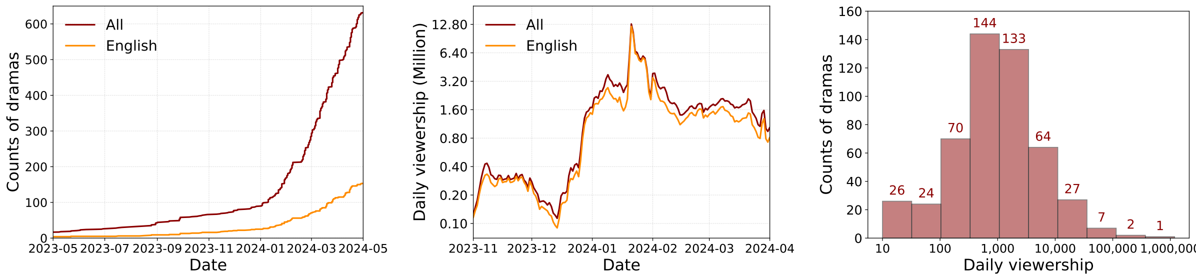


(a) Number of users (b) Geographical distribution (c) Cumulative distribution of age

Figure B.1: Users on the platform

Notes: Panel (a) shows the evolution of the accumulative number of users (left, red) and the daily active users (right, orange) between May 2023 and May 2024. My sample period is highlighted by shaded areas. Panel (b) reports the six countries with the highest revenues and most users up to May 8, 2024. Panel (c) displays the cumulative distribution function of age for new users who entered between March 1, 2024, and May 14, 2024. These aggregate data are directly provided by the platform.

Drama. I report the number and viewership of dramas in Figure B.2. Figure B.2a shows the evolution of the number of drama-language pairs over time. Over my sample period, the total number of drama-episode pairs increased from 67 in November 1, 2023, to 461 in March 31, 2024, where the number of English dramas increased enormously from 16 to 114. The corresponding viewership trends are displayed in Figure B.2b. Consistent with the geographical distribution in Figure B.1b, 83% of viewership during my sample period is in English. The daily viewership is 1.7 million episodes on average, which increased drastically in January 2024 due to the introduction of one "superstar" drama. I further report the distribution of daily viewership in Figure B.2c, which showcases a remarkable heterogeneity between dramas. The superstar drama, which was introduced on January 19, 2024, has about 1 million daily viewership, whereas most dramas have viewerships below 10,000 per day.



(a) Number of dramas (b) Aggregate viewership (c) Viewership distribution

Figure B.2: Drama entry and viewership

Notes: Panel (a) shows the evolution of the number of drama-language pairs on this platform over time. The shaded area indicates for the sample period for my study. Panel (b) reports the aggregate viewership of dramas over my sample period for all dramas and English dramas. Panel (c) plots the distribution of viewership on drama-language level. Note that viewership is displayed in log scale in both the y-axis of panel (b) and x-axis of panel (c).

Revenue. I present the time-series sequence of the three revenue sources in Figure B.3. The main source of all time is the token income, whereas the revenue from subscription has been increasing quickly since February 2024. The ad revenue remains at a relatively low level for the whole sample period.

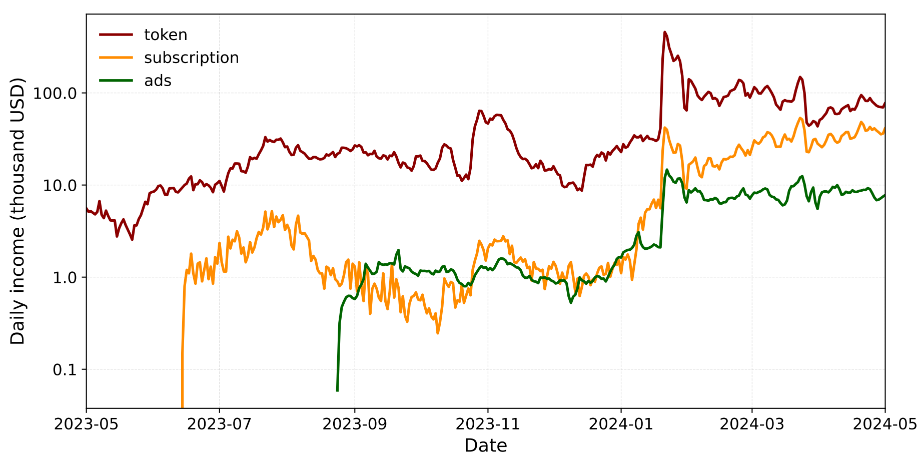
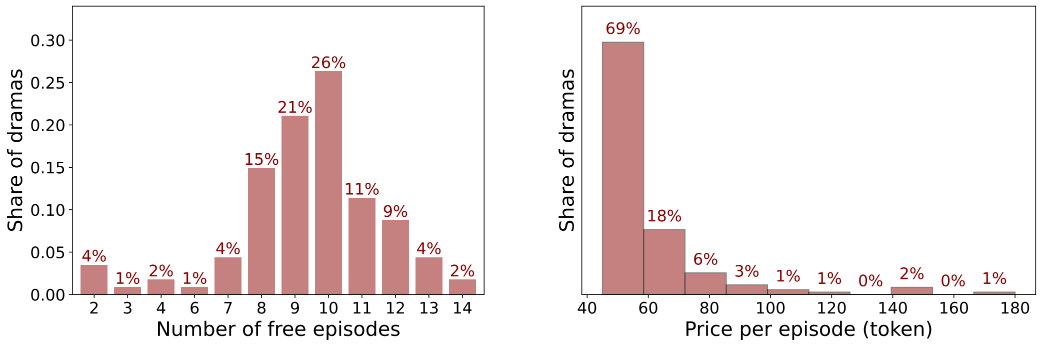


Figure B.3: Three sources of revenue: Token, subscription, and ads

Notes: This figure shows the evolution of different daily revenue sources. My data for subscription and ads started in June 2023 and August 2023, respectively. The level of revenue is displayed in log scale on the y-axis.

Drama pricing. In Figure B.4a, I report the length distribution of free episodes for each drama. Most dramas have eight to 10 free episodes to attract users, whereas the minimum and maximum numbers are two and 14. Figure B.4b displays the price distribution of episodes by dramas in terms of platform tokens. Within a drama, each episode costs the same amount of tokens to unlock, whereas price varies substantially across dramas. Most drama episodes are priced between 45 and 80 tokens.



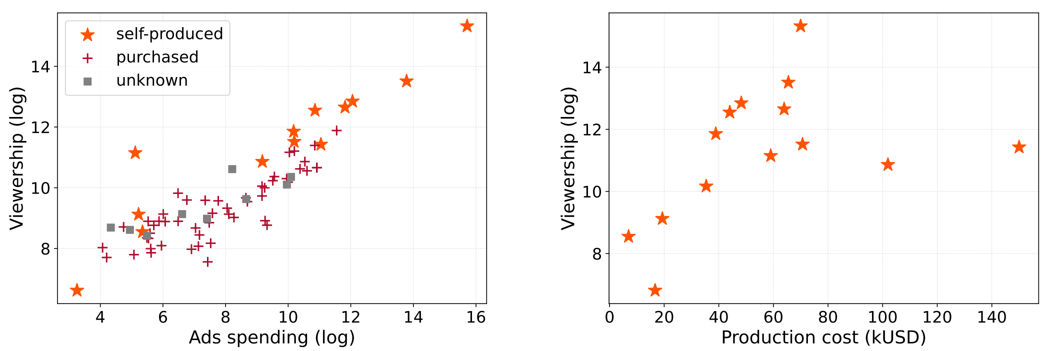
(a) Length of free episodes

(b) Price per episode

Figure B.4: Drama prices and token purchase

Notes: Panel (a) shows the distribution of the number of free episodes by dramas for all English dramas in my sample. Panel (b) plots the distribution of price per episode for those dramas.

Drama supply. In Figure B.5a, I show a positive relationship between ads spending and total viewership on drama level. Consistent with the aggregate description, self-produced dramas in general have higher ads spending and viewership. Conditional on ads spending, self-produced dramas also have higher viewerships than purchased ones, which indicates for some unobserved quality differences. I also observe the total production costs for platform-owned dramas, which are plotted in Figure B.5b against their total viewership. The production costs for dramas are fairly low, because most dramas are produced with costs below $80,000. In general, dramas that cost more are more popular among users, highlighting the need to consider heterogenous quality among different dramas.



(a) Ads spending

(b) Production cost

Figure B.5: Drama ownership, ads spending, and production cost

Notes: Panel (a) plots the total viewership and total ads spending between November 1, 2023, and March 31, 2024, for each drama. The plot distinguishes three types of ownership: self-produced (orange, star), purchased from independent producers (red, square), and dramas whose ownership is unknown to me (grey, circle). In panel (b), I plot the production costs for self-produced dramas against their total viewership.

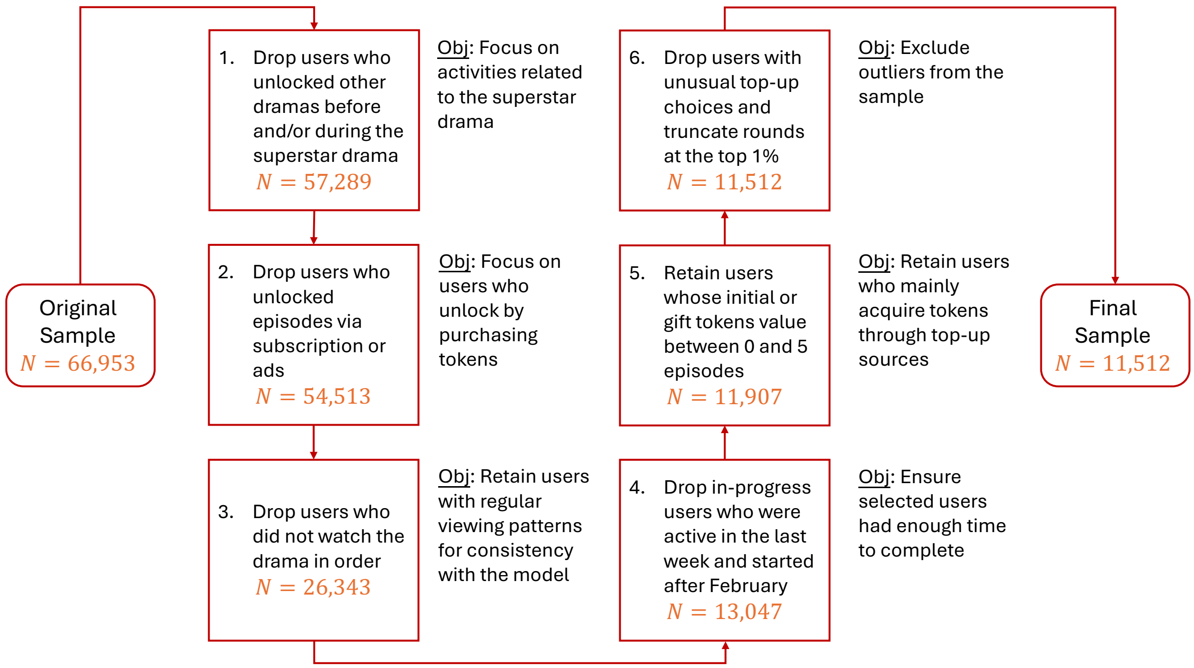


Figure C.1: Data cleaning procedure

Notes: This figure summarizes the data cleaning procedure step by step in the form of a diagram. Each box shows the main filtering step, the resulting sample size (in orange), and the objective of the step (in text alongside).

Appendix C Empirics

Appendix C.1 Data cleaning procedure

This section provides a detailed discussion of the data cleaning procedure. The full process is summarized in six steps in Figure C.1, with each step accompanied by the resulting sample size and its underlying rationale. I elaborate on each step below.

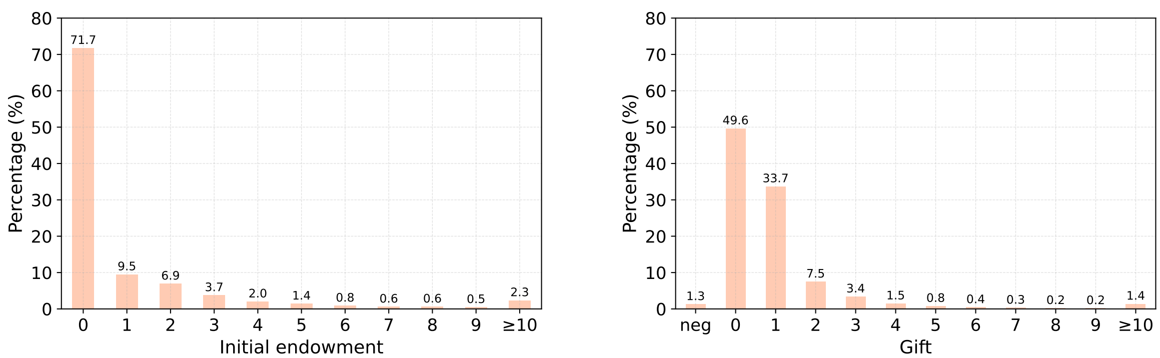
1. *Drop users who unlocked other dramas before and/or during the superstar drama.* This step excludes users with prior experience watching other dramas on the platform, thereby removing potential confounds from pre-existing habit formation. It also ensures that all top-up activity is attributable solely to the superstar drama, aligning the sample with the structural model, which focuses exclusively on that title. The resulting sample remains broadly representative, retaining 85.6% of users, suggesting that this step is unlikely to introduce substantial selection bias.

2. *Drop users who unlocked episodes via subscription or ads.* This step ensures that sample users unlock episodes exclusively through token purchases, which is the primary focus of the analysis. As the vast majority of users (95.2%) use tokens only, this restriction is unlikely to introduce significant selection bias.

3. *Drop users who did not watch the drama in order.* This step retains users with regular, sequential viewing patterns, consistent with the structural model's assumption of episode-by-episode decision-making. It simplifies the modeling exercise by abstracting from the choice of viewing sequence. The cost is a substantial reduction in sample size, with 51.7% of users excluded. If users who watch out of order are less "rational" or more erratic, this selection likely results in a sample of more rational users—potentially leading to an underestimation of self-control problems, thereby reinforcing the paper's core argument.

4. *Drop in-progress users who were active in the last week (March 24–31, 2024) and those who started after February, 2024.* This step ensures that users in the sample had sufficient time to complete the drama, so that incomplete viewing reflects a choice not to continue. While this restriction reduces the sample size by 49.5%, it is unlikely to introduce systematic selection, because the timing of when users begin watching is plausibly random.

5. *Retain users whose initial or gift token value falls between 0 and 5 episodes.* This step excludes users who received a substantial number of tokens from non–top-up sources, such as daily log-ins, which appear as large initial endowments or gift token balances. Figure C.2 shows the distribution of these token values among users entering this step, indicating that most users receive minimal amounts from such sources. This restriction removes only 8.7% of users, suggesting that it is unlikely to introduce significant selection bias.



(a) Distribution of initial endowment

(b) Distribution of gift tokens

Figure C.2: Distribution of initial and gift tokens

Notes: Panels (a) and (b) report the distributions of initial endowments and gift tokens, respectively, among users entering Step 5 of the data cleaning procedure in Figure C.1. Token amounts are normalized by the number of tokens required to unlock one episode.

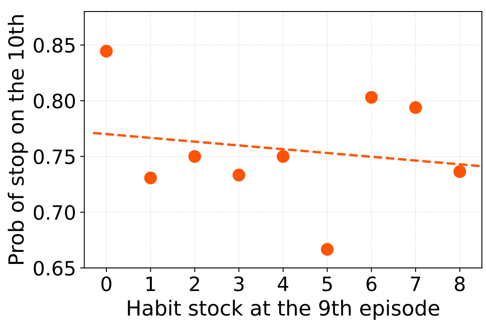
6. *Drop users with unusual top-up choices and truncate rounds at the top percentile.* This step removes outliers from the sample. Unusual top-up packages are defined as those with non-standard prices (i.e., outside the four main packages analyzed in this paper) or token amounts with conditional frequencies below 5%. These packages typically originate from limited-time promotions or pricing experiments conducted by the platform. I also exclude users in the top percentile of the round distribution to eliminate extreme viewing patterns. This step

introduces minimal selection, removing only 3.3% of users.

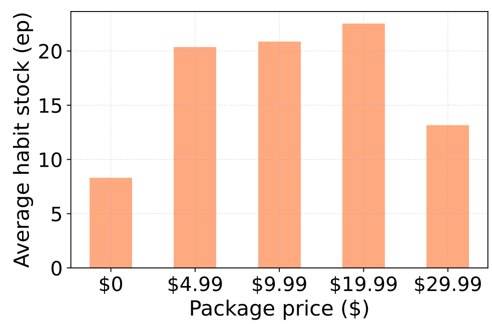
Appendix C.2 Rational addiction

In Figure C.3, I provide suggestive evidence of habit formation in users' drama-watching behavior, a channel commonly interpreted as rational addiction (Becker and Murphy, 1988). Panel C.3a shows the fitted regression described in Section 3.4. Among users who have to top up before unlocking the 10^{th} episode of the superstar drama, a negative relationship exists between the likelihood of stopping and the corresponding habit stock. Those with one additional unit of habit stock are, on average, 0.3% more likely to continue watching.

Panel C.3b shows the average habit stock conditional on selecting each top-up package. A higher-priced package is in general selected by users with higher habit stock, supporting the habit-formation effect. The drop for the $29.99 package is mechanical, because users no longer need the largest package when they have already watched most episodes, and few remain to unlock.



(a) Conditional probability to stop



(b) Top-up package choice

Figure C.3: Suggestive evidence on rational addiction (habit formation)

Notes: Panel (a) shows that, among users who have to top up before unlocking the 10^{th} episode of the superstar drama, a negative relationship exists between the likelihood of stopping and the corresponding habit stock. Those with one additional unit of habit stock are, on average, 0.3% more likely to continue watching. Panel (b) presents the average habit stock by top-up package choice, where higher-priced packages are generally chosen by users with higher habit stock, consistent with habit formation. The drop for the $29.99 package is mechanical, because users no longer need the largest package when they have already watched most episodes and few remain to unlock.

Appendix C.3 Viewership conditional on time of a day

Figure C.4 reports the number of unlockings conditional on hour of a day (Eastern time). It suggests the usage between 1am and 10am is below average, depending on which I define the "night-time" outside option for my structural analysis.

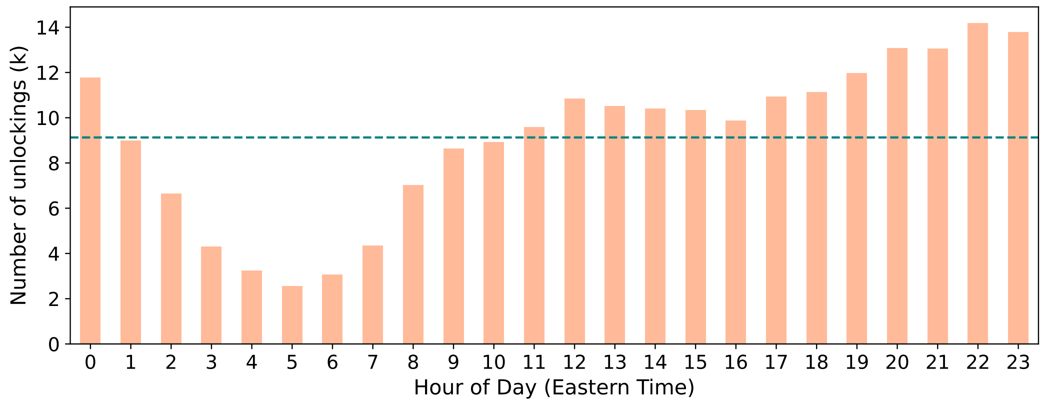


Figure C.4: Usage conditional on time of a day

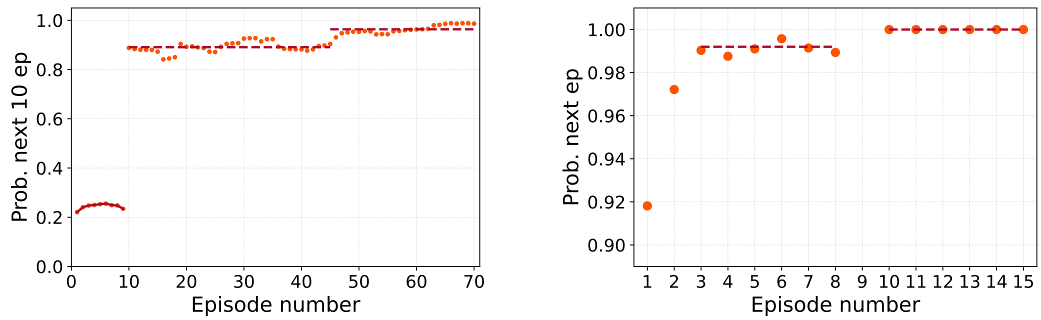
Notes: This figure represents the total number of unlocking activities that happened in each hour of a day within my sample. The hour is based on Eastern time. The teal, dashed line is the average number of unlocking per hour.

Appendix C.4 Other rational motives

In this section, I provide additional evidence that learning effects and other rational motives alone do not appear sufficient to explain users' drama-watching behavior. Two observations support this interpretation. First, the conditional probability of continuing to watch stabilizes within one or two episodes, suggesting that learning happens quickly. Second, I provide conservative payment evidence for overpayment from the perspective of package choices.

Evidence 1: The probability of continuing watching stabilizes quickly. In Figure C.5a, I present the evolution of the probability of continuing to the next 10 episodes, conditional on each episode. This probability initially rises from 22.1% at the first episode to 24.8% by the third, and then stabilizes around 25% until the ninth episode. A sharp increase occurs at the 10th episode—when users must start paying for new episodes—pushing the probability to 89.1%, where it remains stable between episodes 10 and 45. As the show nears its conclusion (episodes 45–70), the probability of continuing rises further to 96.3%. This analysis yields two key insights. First, the conditional probability increases significantly only during the first two episodes, after which it stabilizes quickly, suggesting learning occurs rapidly. Second, the price effect at the 10th episode is substantial, causing a 64.1-percentage-point increase in the continuation probability, whereas the learning-related movement accounts for only a few percentage points overall. This pattern suggests that slow learning is unlikely to explain the repeated paid top-up margin by itself.

These insights are further confirmed in Figure C.5b, which illustrates the evolution of the conditional probability of continuing to the next episode. Initially, this probability is relatively low, at 91.8% for the first episode and 97.2% for the second. From the third episode onward, it stabilizes around 99%, indicating any learning effect occurs primarily within the first two episodes.



(a) Next 10 episodes

(b) Next episode

Figure C.5: Conditional probability of watching the next few episodes

Notes: This set of figures presents the probability that a user will continue watching 10 (panel (a)) or one (panel (b)) conditional on watching each episode. The orange circles are the conditional probability, whereas the red, dashed lines provide fitted values. For a clean comparison, the conditional probability is calculated among users with zero initial token endowment.

Evidence 2: Purchase-time payment evidence. I next report a package-replacement exercise that provides a conservative lower bound on purchase-time overpayment associated with repeated small packages. Following the payment logic in DellaVigna and Malmendier (2006), I ask whether users who purchased the $4.99 package in a given episode bin would have spent less if that purchase had instead been replaced by a larger package available in the same menu. The exercise is evaluated at the purchase time: it asks whether a cheaper commitment was available at that decision point, while holding the user's subsequent realized viewing fixed.

This calculation is deliberately conservative. First, it does not classify all small-package purchases as mistakes. Small packages may reflect learning, uncertainty about future outside options, unused-token concerns, or other rational reasons for choosing low commitment. These cases remain in the residual. Second, after the counterfactual replacement, I do not allow users to reoptimize later top-up decisions; subsequent purchases are held fixed or mechanically adjusted, and unused tokens are credited at the user's average token price. Allowing users to reoptimize later purchases would weakly increase the savings from the replacement. Positive entries in Table C.1 should therefore be interpreted as lower-bound payment evidence of ex-ante overpayment under the package-choice interpretation, rather than as a structural estimate of the total self-control loss.

The key takeaway from Table C.1 is that the replacement exercise generates positive average savings for some late $4.99 purchases even under this conservative accounting. Replacing the first $4.99 package with the $9.99 package saves $0.20, $0.30, and $2.62 per user on average in the 40–50, 50–60, and 60–70 episode bins, respectively. The larger-package replacements impose stronger commitment and are therefore noisier, but the $19.99 replacement also generates positive average savings in the 40–50 bin. Negative entries should not be read as evidence against self-control problems: users may rationally choose the $4.99 package to learn about quality, preserve flexibility over outside options, or avoid unused tokens, and the exercise does not allow reoptimization after the replacement. The positive late-bin entries therefore provide conservative payment evidence

Table C.1: Purchase-time package replacement: Ex ante savings per user from replacing the $4.99 package

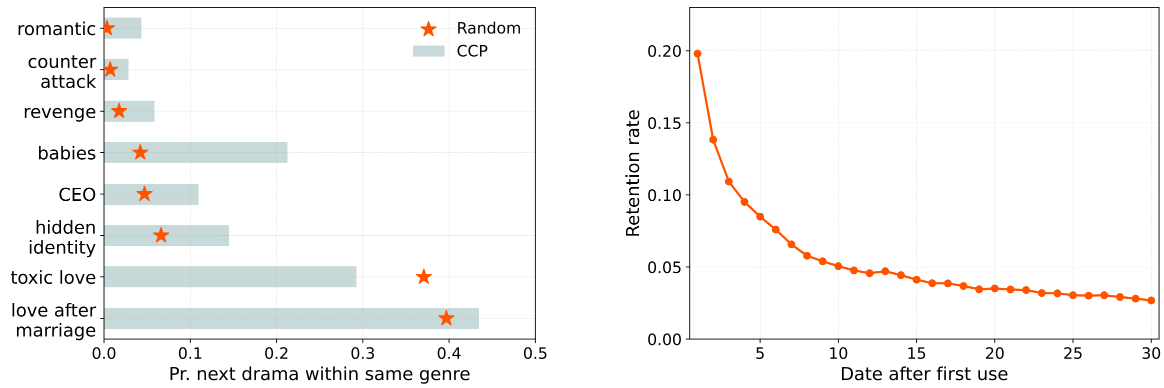
Episode (bin)		10-20	20-30	30-40	40-50	50-60	60-70	70-80
Replace $4.99 with:	$9.99	–1.72	–0.72	–0.84	0.20	0.30	2.62	–3.28
	$19.99	–6.71	–4.75	–4.04	1.75	–2.05	–6.16	–13.28
	$29.99	–10.05	–8.52	–8.70	–7.65	–12.05	–16.16	–23.28
Number of observations		797	203	140	97	74	83	154

Notes: In this exercise, I hold fixed each user's total number of episodes and replace their first $4.99 purchase within a given episode bin with a larger token package ($9.99, $19.99, or $29.99). All other top-ups are kept at their original choices before the switch or approximated by the closest original decision after the switch. To account for the discreteness of token purchases, I adjust unused tokens from additional packages by imputing their value at the user's average token price. The table reports average net savings per user, where negative values indicate higher costs. Positive entries are lower-bound payment evidence of ex-ante overpayment under the package-choice interpretation: they survive rational motives for low-commitment purchases and a conservative accounting that does not allow users to reoptimize after the replacement. Negative entries are treated as ambiguous rather than as evidence against self-control problems.

that repeated small-package purchases become costly after substantial viewing experience.

Appendix C.5 No spillover across dramas

I find limited evidence of spillovers beyond the focal drama, which supports analyzing the single superstar drama as the main consumption environment. I check history dependence at both the genre and platform levels. In Figure C.6a, I report the conditional choice probability (CCP) that a user who has just finished one drama watches another drama in the same genre (the teal bar), along with the probability from random selection based on the empirical distribution (the orange star). The results show no sizable difference between the two; if anything, user selection would mechanically make the CCP larger than random choice. At the platform level, Figure C.6b plots retention rates, defined as the fraction of users who are still active X days after first use. The 30-day retention rate is 2.7%, which is low relative to other digital platforms. These patterns suggest that the main persistence to be explained occurs within the focal drama rather than through broad platform-level state dependence.



(a) No spillover within a genre

(b) No spillover within the platform

Figure C.6: No spillover across dramas

Notes: The teal bars in panel (a) show that, conditional on users just finishing a drama with the corresponding genre, the probability that the next drama they watch will share the same genre. The orange star indicates the probability if the users randomly select the next drama to watch based on the empirical distribution. Panel (b) plots the retention rate on the platform with varying number of days, which is the fraction of users who are still active X days after their first use. The 30-day retention rate is 2.7%, which is lower than other online platforms such as TikTok (30%), Netflix (14.4%), and GoodNovel (7.5%).

Appendix D Structural Model

Appendix D.1 Solution characterization

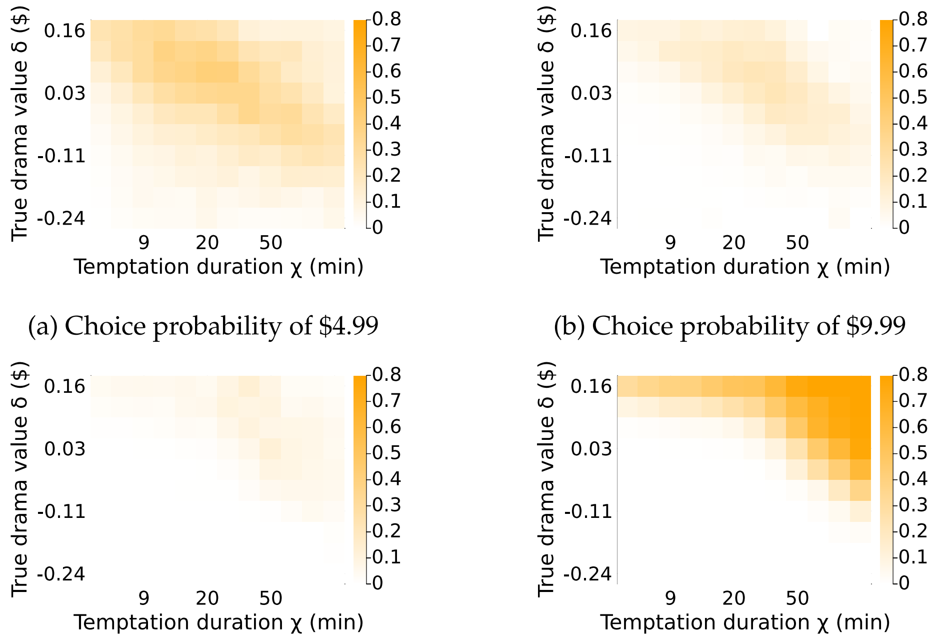
Top-up package choices. With the full solution in hand, I can characterize the conditions under which users purchase each top-up package. For intuition, consider the case without package taste shifters ($\nu_k \equiv 0$). When $t > 9$ and the token balance is $a_t = 0 < c_t$, the perceived value of top-up package k for a realized posterior mean $\widehat{\delta}_t$ is given by:

$$\pi_{kt}(h; \widehat{\delta}_t, \psi, \chi) = EW_t(h, b_k; \widehat{\delta}_t, \psi, \chi) - \omega p_k. \tag{D.1}$$

The top-up choice, therefore, depends on the shape of $EW_t(h, b_k; \widehat{\delta}_t, \psi, \chi)$, which is increasing in continuation value, posterior quality, habit stock, and effective temptation duration. Figure D.7 plots the estimated choice probability for each package by latent value and temptation-duration bins, averaging over the simulated belief paths generated by the learning model.

The outside option ($k = 0$) yields a value of zero and will be selected if $\pi_{kt}(h; \widehat{\delta}_t, \psi, \chi) < 0$ for all $k \in \{1, \ldots, K\}$. This typically occurs when the user's posterior value is sufficiently low. Because simulated users with low latent δ tend to hold lower posterior means after the early episodes, the lower-value region in Figure D.7 has low purchase probabilities for all packages.

At the other extreme, the largest package becomes most attractive when posterior value and continuation value are high enough that the user anticipates watching many paid episodes. In that region, the package choice resembles cost minimization over expected future unlocks: the \$29.99 package offers the lowest price per token and is therefore chosen early in the paid portion



(a) Choice probability of \$4.99

(b) Choice probability of \$9.99

(c) Choice probability of \$19.99

(d) Choice probability of \$29.99

Figure D.7: Choice probability of each top-up package between episodes 10 and 15

Notes: The four heat maps report the estimated choice probability of each top-up package between episodes 10 and 15 conditional on latent value δ and effective temptation duration χ. The policy is solved over posterior beliefs; these figures bin users by latent value for visualization and average over the simulated learning paths. The warmer the color, the larger the choice probability. The probabilities from the four figures do not sum to 1, with the gap being the share that choose the outside option ($k = 0$).

of the series. This intuition is supported by Figure D.7d, where high-value users are much more likely to purchase the largest package between episodes 10 and 15.

The smaller and medium-priced packages are most informative for the temptation-duration distribution. Holding posterior value fixed, a longer effective temptation duration raises the number of episodes the user expects to continue watching, increasing the value of packages with more tokens. At the same time, repeated returns to the smallest package remain informative about renewed short-run temptation. The package-choice surface therefore helps separate the intensive magnitude of temptation, κ, from heterogeneity in the duration of effective rolling temptation, χ.

Unlocking choices. With sufficient tokens, the user unlocks an episode if and only if the perceived value of unlocking exceeds the outside option, namely $u_t(h, \widehat{\delta}_t, \psi, \chi, \epsilon) + \mathcal{C}_t(h, a; \widehat{\delta}_t, \psi, \chi) > 0$. Because both flow utility and continuation value are increasing in posterior value, habit stock, and effective temptation duration, the unlocking decision follows a threshold rule in these state variables. For identification, unlocking dynamics therefore provide information on the relative magnitudes of intrinsic value, learning, habit formation, and temptation.

Table D.2: Two-stage estimation procedure

	Parameter												
	μ_δ	σ_δ	σ_η	$\overline{\psi}$	β_ϵ	κ	α_χ	π_0	α_1	α_2	ρ	ω	obj.
First stage	-0.0202	0.0931	0.0389	0.0191	0.0462	0.211	0.515	0.285	-1.40e-5	1105.74	0.0461	0.905	2762.62
Second stage	-0.0202	0.0928	0.0386	0.0191	0.0462	0.211	0.516	0.288	-1.40e-5	1103.87	0.0463	0.902	2755.80

Appendix D.2 Estimation algorithm

This appendix reports the targeted moments and summarizes the two-step estimation routine used for the active-learning specification. For each candidate parameter vector $\boldsymbol{\theta}$, I calculate model moments by simulating the economy 20 times and taking the average. To reduce sensitivity to starting values and local optima, I apply a two-step procedure that combines a grid search with a bounded local solver for the SMM optimization problem (17).

Global grid search. I first conduct a global search over the current 12-dimensional parameter vector. I evaluate the SMM objective on each grid point and select the on-grid minimizer. The best grid point has an objective value of 2762.62 and yields the first-stage estimates reported in Table D.2.

Local solver: Nelder-Mead algorithm. I then refine the first-step global minimizer $\widehat{\boldsymbol{\theta}}^{(1)}_{SMM}$ with a bounded local solver. The local search uses normalized coordinates within the grid box and applies the gradient-free Nelder–Mead algorithm from `Optim.jl`. This step improves the objective value from 2762.62 to 2755.80.

Appendix D.3 Targeted moments

I report the targeted moments in Table D.3. I normalize the habit stock by the total number of episodes (80), top-up user share by the number of users in each episode bin, and unlocking share by the total number of users. The learning-specific rows report transition probabilities after the first and repeated $4.99 purchases, together with all-user tail probabilities for repeated paid top-ups. The model column is updated from the solver-side moment export; displayed values are rounded to three decimals.

Table D.3: Targeted moments

No	Moment	Note	Data	Model
1	`E_ccp_j10_h_leq_4`	Mean of stopping CCP for the 10th ep, conditional on last $h \leq 4$ and no endowment	0.762	0.796
2	`E_ccp_j10_h_5_8`	Mean of stopping CCP for the 10th ep, conditional on $5 \leq h \leq 8$ and no endowment	0.750	0.746
3	`E_h_k_0`	Mean of last habit stock when user chooses $k = 0$, normalized by T	0.091	0.101
4	`E_h_k_1`	Mean of last habit stock when user chooses $k = 1$	0.253	0.266

No	Moment	Note	Data	Model
5	E_h_k_2	Mean of last habit stock when user chooses $k = 2$	0.274	0.277
6	E_h_k_3	Mean of last habit stock when user chooses $k = 3$	0.299	0.180
7	E_h_k_4	Mean of last habit stock when user chooses $k = 4$	0.165	0.097
8	q_k1_j_10_20	User share of package $k = 1$ sold at $t \in [10, 20]$	0.337	0.508
9	q_k1_j_21_40	User share of package $k = 1$ sold at $t \in [21, 40]$	0.179	0.300
10	q_k1_j_41_60	User share of package $k = 1$ sold at $t \in [41, 60]$	0.122	0.267
11	q_k1_j_61_80	User share of package $k = 1$ sold at $t \in [61, 80]$	0.187	0.259
12	q_k2_j_10_20	User share of package $k = 2$ sold at $t \in [10, 20]$	0.292	0.161
13	q_k2_j_21_40	User share of package $k = 2$ sold at $t \in [21, 40]$	0.231	0.144
14	q_k2_j_41_60	User share of package $k = 2$ sold at $t \in [41, 60]$	0.214	0.131
15	q_k2_j_61_80	User share of package $k = 2$ sold at $t \in [61, 80]$	0.170	0.056
16	q_k3_j_10_20	User share of package $k = 3$ sold at $t \in [10, 20]$	0.155	0.055
17	q_k3_j_21_40	User share of package $k = 3$ sold at $t \in [21, 40]$	0.091	0.039
18	q_k3_j_41_60	User share of package $k = 3$ sold at $t \in [41, 60]$	0.206	0.010
19	q_k3_j_61_80	User share of package $k = 3$ sold at $t \in [61, 80]$	0.074	0.000
20	q_k4_j_10_20	User share of package $k = 4$ sold at $t \in [10, 20]$	0.154	0.316
21	q_k4_j_21_40	User share of package $k = 4$ sold at $t \in [21, 40]$	0.023	0.003
22	q_k4_j_41_60	User share of package $k = 4$ sold at $t \in [41, 60]$	0.039	0.000
23	q_k4_j_61_80	User share of package $k = 4$ sold at $t \in [61, 80]$	0.010	0.000
24	unlock_2	Share of unlockings at ep 2	0.937	0.946
25	unlock_3	Share of unlockings at ep 3	0.917	0.928
26	unlock_4_9	Share of unlockings at ep 4–9	0.894	0.908
27	unlock_10_14	Share of unlockings at ep 10–14	0.279	0.289
28	unlock_15_80	Share of unlockings at ep 15–80	0.143	0.148
29	unlock_night	Fraction of unlockings in the night	0.264	0.268
30	first_k1_stop	Next paid top-up outcome after first-ever $k = 1$: no next paid top-up	0.384	0.433
31	first_k1_next_k1	Next paid top-up outcome after first-ever $k = 1$: $k = 1$	0.298	0.341
32	first_k1_next_k2	Next paid top-up outcome after first-ever $k = 1$: $k = 2$	0.240	0.150
33	first_k1_next_k3	Next paid top-up outcome after first-ever $k = 1$: $k = 3$	0.060	0.041
34	first_k1_next_k4	Next paid top-up outcome after first-ever $k = 1$: $k = 4$	0.017	0.035
35	repeat_k1_j12	Next paid top-up is $k = 1$, conditional on current $k = 1$ and paid order $j \in \{1, 2\}$	0.320	0.363
36	upgrade_k1_j12	Next paid top-up is $k \in \{2, 3, 4\}$, conditional on current $k = 1$ and paid order $j \in \{1, 2\}$	0.320	0.236
37	repeat_k1_j345	Next paid top-up is $k = 1$, conditional on current $k = 1$ and paid order $j \in \{3, 4, 5\}$	0.447	0.468
38	upgrade_k1_j345	Next paid top-up is $k \in \{2, 3, 4\}$, conditional on current $k = 1$ and paid order $j \in \{3, 4, 5\}$	0.224	0.222
39	repeat_topup_ge3_all_users	All-user share with at least 3 paid top-ups; non-payers count as zero	0.067	0.058
40	repeat_topup_ge5_all_users	All-user share with at least 5 paid top-ups; non-payers count as zero	0.021	0.025

Appendix D.4 Decomposition of perceived utility

The estimated model separates welfare-relevant habit formation from choice-utility temptation. I decompose perceived flow utility into three components: intrinsic value, temptation, and habit formation. Figure D.8 displays the average value of each component for users who have watched

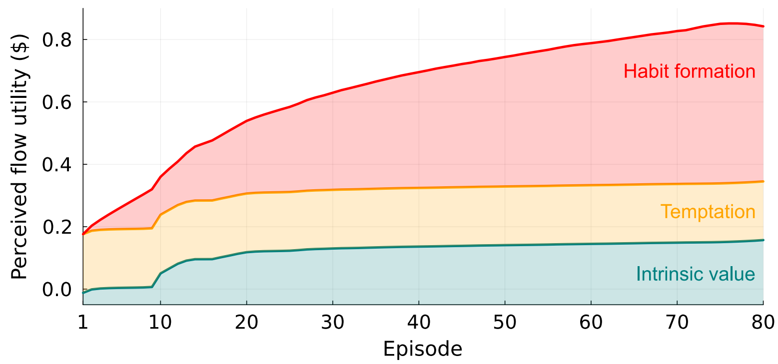


Figure D.8: Decomposition of perceived flow utility by episode

Notes: This figure decomposes the perceived flow utility into three components by episode: intrinsic value, temptation, and habit formation. For each episode, I report the average level of each utility component conditional on unlocking. All the components are normalized in dollars.

episode t. The teal area represents the average intrinsic value per episode, increasing from -1.2¢ for the first episode to 16¢ for the last, reflecting user selection over episodes as low-valuation users gradually stop watching. The sharp rise between episodes 10 and 14 is due to an exogenous increase in episode prices from 0 to 1 token, leading over three-quarters of users to stop watching. On average, users derive $1.43 intrinsic utility from this drama relative to their outside options.

The estimated temptation component is substantial relative to intrinsic value. The orange area represents the value of temptation, valued at 19¢ per one-minute episode and providing an average perceived value of $3.67 per user from the drama series. This component affects choice utility but is excluded from the long-run welfare criterion. The red area at the top reflects the average level of habit formation for each episode, which gradually increases as users build habit stock over episodes. For most episodes, habit formation is the primary component of utility, helping explain users' high willingness to pay conditional on paying. However, the majority of users stop watching between episodes 10 and 15, which reduces the average impact of habit formation across users. On average, users derive $4.30 from habit formation throughout the series, highlighting the empirical importance of separating welfare-relevant habit formation from choice-utility temptation.

Appendix E Robustness of the Short-Video Extrapolation

This appendix summarizes the robustness exercises for the short-video extrapolation in Section 7. The full-transfer benchmark imports the estimated distribution of net intrinsic value $x = \delta - \psi$, the temptation magnitude κ, and the temptation-horizon distribution χ into a one-hour short-video environment with one-minute content units. The benchmark should be interpreted as a scale calculation, not as an estimate of welfare on TikTok or any specific platform. I therefore report several sensitivity exercises that relax the strongest transfer assumptions.

Table E.4 summarizes the main exercises. The first row reports the full-transfer benchmark. The next rows relax, in turn, the value distribution, the share of users assumed to be behaviorally comparable to short-drama users, the transfer of temptation magnitude and duration, and the compliance assumption for the default time limit. I also report a conservative combined stress test that weakens multiple assumptions simultaneously. Across these exercises, the implied welfare losses and policy gains remain economically meaningful.

The full-transfer benchmark implies a temptation-induced loss of \$2.16 per hour and \$13.3 billion per month. Increasing the content unit to 12 minutes recovers \$7.9 billion per month, while a 10-minute default time limit recovers \$7.8 billion. These large magnitudes rely on strong transfer assumptions. The remaining rows show how the conclusions change when those assumptions are relaxed. If only half of U.S. TikTok users are behaviorally comparable to short-drama users, the implied monthly loss falls to \$6.6 billion and the 10-minute default gain falls to \$3.9 billion. If both temptation magnitude and temptation duration are scaled down by one half, the monthly loss remains \$5.6 billion and the default gain remains \$3.3 billion. Combining these restrictions gives the conservative stress test: the monthly loss is \$2.8 billion and the default gain is \$1.7 billion.

The table also separates welfare sensitivity from policy-implementation sensitivity. Under the full-transfer benchmark, the 10-minute default recovers \$7.8 billion per month when all would-be compliers comply. If only half comply, the gain falls to \$3.9 billion. This exercise is useful because real-world default effects depend on salience, interface design, and opt-out frictions. The policy conclusion therefore does not require full compliance: even partial compliance generates sizable gains under the transferred welfare criterion.

Table E.4: Short-video extrapolation: Robustness and sensitivity

Scenario	Assumption / implementation	Loss/hr	Loss, B/mo	$n = 12$ gain, B/mo	10-min default gain, B/mo
Full transfer	Transfers x, κ, and χ from the short-drama estimates; all users are treated as comparable.	\$2.16	\$13.3	\$7.9	\$7.8
Non-user-share matched	Shifts the value distribution by -0.130 to match a 47.2% non-user share among U.S. internet users. This is a diagnostic, not a conservative case.	\$3.93	\$36.9	\$24.4	\$21.8
Partial population transfer	Only $q = 0.50$ of users are behaviorally comparable to short-drama users; the remaining users have zero temptation-induced loss.	\$2.16	\$6.6	\$4.0	\$3.9
Weaker temptation transfer	Scales both temptation magnitude and temptation horizon by one half: $\lambda_\kappa = \lambda_\chi = 0.50$.	\$0.68	\$5.6	\$3.6	\$3.3
Conservative combined stress test	Combines $q = 0.50$ with $\lambda_\kappa = \lambda_\chi = 0.50$. This is a stress test, not a formal lower bound.	\$0.68	\$2.8	\$1.8	\$1.7
Imperfect default compliance	Keeps the full-transfer benchmark but assumes only 50% of would-be compliers follow the 10-minute default.	\$2.16	\$13.3	–	\$3.9

Notes: The table reports robustness exercises for the short-video extrapolation. "Loss/hr" is the temptation-induced surplus loss per hour under the transferred long-run welfare criterion. Monthly aggregates use 170 million U.S. TikTok users and 29.2 monthly hours. For rows involving partial population transfer, aggregate quantities are scaled by the comparable-user share q. The $n = 12$ column reports the aggregate monthly gain from increasing the content unit to 12 minutes. The final column reports the aggregate monthly gain from a 10-minute default time limit. The non-user-share-matched row shifts the value distribution to match the extensive margin of TikTok adoption and should be interpreted as an external-validity diagnostic rather than a conservative estimate.